Opening the Black Box*

A Theory of the Value of Data

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Latest Draft

Abstract

This paper develops a theory of the value of data for prediction. An agent observes an outcome and some characteristics (covariates) of a sample of individuals to predict the outcome for a target individual based on her characteristics. The main findings are: (i) covariates exhibit economies of scope, as the value of one covariate is higher when others are also observed; (ii) covariates and observations are complements when data are scarce but become substitutes when data are abundant. These findings have three policy implications. Mergers between firms having different covariates can be privately profitable yet reduce welfare, especially when data are scarce (e.g., under strict privacy rules). Allowing firms to pool covariates is always procompetitive because it removes double marginalization, whereas pooling observations can be anticompetitive when data are abundant. Finally, a data seller may profitably exclude one of several competing prediction providers even when this lowers total welfare.

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1 Introduction

In digital markets, data drive competitive advantage because they allow firms to make better predictions: ChatGPT and Gemini predict the next response a user is likely to find helpful given a prompt and context; Amazon and Uber predict where demand will arise; Google and Meta predict which ad a user will click; and Netflix predicts what content a user will enjoy most. Over the last decade, many digital markets have become dominated by a few firms that control vast amounts of data, attracting the attention of competition authorities. For instance, according to the 2021 U.S. House Report, "data advantages [...] can reinforce dominance and serve as a barrier to entry,"[1] a warning echoed in the EU Data Act proposal[2] and in major policy reports.[3]

These concerns are partly fueled by earlier statements from industry leaders: as Google's CEO declared in 2009, "Scale is the key. We just have so much scale in terms of the data we can bring to bear."[4] However, more recently, technology firms have downplayed policymakers' claims, arguing that data exhibit diminishing returns to scale by the Law of Large Numbers, and that data concentration in the hands of few firms reflects their technological advantage on rivals rather than entry barriers.[5]

To evaluate these competing narratives, we must consider two distinct ways in which firms acquire data: either by observing more individuals (i.e., increasing sample size) or by observing more attributes of each individual (i.e., increasing the covariates). This paper asks a simple but unresolved question: how do predictions improve when a firm acquires (i) more observations and/or (ii) more covariates? Empirically and conceptually, these two channels—*economies of scale* and *economies of scope*—are often conflated. My goal is to disentangle them.

I develop a Bayesian model that cleanly disentangles these two dimensions. A firm predicts an outcome for a target individual—for example, the price of a house. The outcome depends on many potential covariates (observable attributes of the house, e.g., size, number of rooms, year of construction), each of which has an unknown impact on the outcome. The firm begins with prior uncertainty about these impacts and can collect two types of data: (1) *Training data*: covariates and outcomes for a sample of individuals, used to learn the relationship between covariates and outcomes; (2) *Prediction covariates*: a subset of the covariates of the target individual, used to apply what the firm has learned from the training data.

Consequently, the firm (Zillow) designs a dataset by choosing three objects: (i) the number of observations in the training data N (the number of past house prices), (ii) the set of *training covariates* $\mathcal{T}$ (which house characteristics on past purchases), and (iii) the set of *prediction covariates* $\mathcal{P}$ to observe for the target individual (which characteristics to observe on the house

[1] U.S. House of Representatives (2020), Investigation of Competition in Digital Markets, 117-40, pp. 36–38.
[2] European Commission (2023), Data Act proposal, COM(2023) 193 final.
[3] Digital Platforms (2019) and UK Competition and Markets Authority (2019).
[4] Schmidt (2009), "How Google Plans to Stay Ahead in Search," Bloomberg, 2 October.
[5] Varian (2018) and Bajari et al. (2019).

whose price must be predicted).

Because collecting either type of data is costly, the dataset design choice reflects the trade-off between better predictions and higher data collection costs.

Given any dataset design, I derive the optimal Bayesian predictor and I characterize the value of the data, defined as the expected reduction of out-of-sample mean squared error achieved by observing the data. In this framework, each training covariate improves prediction by reducing uncertainty about its effect, while each prediction covariate allows the firm to apply what it has learned to the target individual. The key quantities are therefore: (i) how much the training data teach the firm about each covariate, and (ii) how much each prediction covariate matters for the outcome.

I show how these components combine into a closed-form expression for the value of any dataset design. The value rises when (i) training covariates reduce residual noise, (ii) additional observations sharpen parameter estimates, and (iii) prediction covariates allow the firm to apply the learned relationship between covariates and outcome to the target individual. I show that there are two regimes depending on whether observations are scarce or abundant. In the observation-scarce regime, the same covariates are used for training and prediction, and covaraites have decreasing marginal returns. Furthermore, covariates and observations are complements.

In the observation-rich regime, the prediction covariates will be a subset of the training covariates (in machine-learning jargon, the firm will "train large deploy small"). In this regime two surprising findings:

First, *training covariates exhibit economies of scope*: adding a new training covariate is more valuable when many other covariates are already observed for the sample. Learning one relationship improves the firm's understanding of all the others because it reduces the residual noise in the regression. This implies there are increasing returns to training covariates.

Second, *training covariates and observations are complements when data are scarce, but become substitutes when data are abundant.* With small datasets, adding a covariate raises the marginal value of an observation; with large datasets, extra covariates mainly reduce residual noise, lowering the marginal value of further observations.

Third, *training data and prediction covariates are complements.* Any improvement in how well the model is trained raises the predictive value of each covariate observed on the target individual, generating positive spillovers from training data to prediction covariates.

I show that these findings generate policy-relevant implications in three applications. First, I study data-driven acquisitions. First, incumbents may acquire entrants even without product or user overlaps, and even when entry by the smaller firm would be socially valuable. Even when firms operate in independent prediction markets, an incumbent with a rich set of covariates (e.g., Google) optimally acquires an entrant with distinct covariates (e.g., Fitbit). This incentive arises purely from economies of scope in covariates: merging datasets raises the marginal value of every covariate and creates a statistical analogue of a natural monopsony

in data.

Second, I study data pools, a type of agreement between firms explicitly encouraged in the EU Data Act and in European common data-space initiatives. In the model, datasets can be fragmented either in observations (firms hold the same covariates on different users) or in covariates (firms hold different covariates on the same users). Pooling along these two dimensions has fundamentally different welfare effects. Pooling covariates on the same individuals combines complementary features and always generates procompetitive efficiencies by eliminating a Cournot-style double marginalization between data brokers. By contrast, pooling observations combines substitute datasets: when data are abundant, competition between brokers is already intense and pooling becomes effectively collusive.

Third, I apply the framework to licensing of training datasets in prediction markets, motivated by recent agreements such as the 2024 Reddit–OpenAI deal. A data seller owns a non-rival training dataset and may license it to two competing prediction firms, each of which can combine the licensed training data with its own proprietary covariates on target users. Licensed and proprietary data are strict complements in generating prediction accuracy, so sellers my raise the buyer's willingness to pay by charging a price so high the rival would not want to buy the license (de facto exclusivity). However, de facto exclusivity also discourages the excluded firm from investing in proprietary data. The model shows that de facto exclusivity is privately profitable whenever competition for prediction buyers is sufficiently fierce, yet it can be socially harmful in an intermediate region where both firms would invest under non-exclusive access, because it precludes investment by the excluded buyer.

Roadmap. Section 2 introduces the data-generating process and the firm's dataset-design problem. Section 3 derives the optimal predictor, highlighting the role of training information and the misspecification penalty. Section 4.2 analyzes the value of a dataset design, develops the core comparative statics on economies of scale and scope, and studies optimal covariate selection under data-collection costs. Section 5 applies the framework to three policy-relevant environments—data-driven acquisitions, data pools, and exclusive licensing. Section 6 concludes. Proofs are collected in Appendix and extensions are in Appendix **??**.

Related Literature. There is a rich information-design literature on the value of data, starting with Bergemann, Bonatti, and Smolin (2018) and continuing with Jones and Tonetti (2020), Bergemann, Bonatti, and Gan (2022), Bergemann and Bonatti (2024), and Acemoğlu et al. (2022). These papers characterize how much an agent is willing to pay for more informative experiments by modeling the choice of an information structure. In contrast, my setting is more structured than a Blackwell experiment and allows me to distinguish observations, training covariates, and prediction covariates. This lets me map the value of data to transparent statistical objects (covariate variances, posterior variances, and a misspecification penalty) and separate the roles of sample size, scope, and covariate selection. Methodologically, the pa-

per is related to Montiel Olea et al. (2022), Iyer and Ke (2024), and Dasaratha, Ortner, and Zhu (2025), who study competition between models with different covariates; here, covariates and observations are modeled jointly and training and prediction data are separated, which generates structural non-convexities and complementarities in the value of data. Finally, Strzalecki (2024) interpret deviations from Bayesian updating via a primitive misspecification parameter; in my framework, an analogous distortion arises endogenously as standard Bayesian behavior under incomplete data collection, providing a microfoundation for likelihood discounting.

I also contribute to the broader literature on economies of scope and scale in data. Most models fix the set of covariates and study returns to observations, as in Bajari et al. (2019) and Goldfarb and Tucker (2011). By endogenizing which covariates are collected, my framework rationalizes empirical evidence on complementarities and economies of scope in Schaefer and Sapi (2023) and Carballa-Smichowski et al. (2025b). Schaefer (2025) develops a complementary frequentist approach and show that the eigenspectrum of the covariate matrix shapes returns to scale, in line with my comparative statics. Allcott et al. (2025) estimate diminishing returns to additional observations and limited complementarities across search queries. While Radner and Stiglitz (1984) attribute increasing returns to information costs, here non-convexities and increasing returns arise purely from the statistical structure of prediction, even under separable reduced-form data-collection costs. These results provide microfoundations for work in IO and macroeconomics that takes increasing returns to data as primitive, either via feedback between data and demand or by assuming complementarities across datasets (Hagiu and Wright 2023; Prüfer and Schottmüller 2021; Farboodi and Veldkamp 2025; Aral, Brynjolfsson, and Wu 2008; Cong, He, and Yu 2021; Carballa-Smichowski et al. 2025a; De Corniere and Taylor 2025; Calzolari, Cheysson, and Rovatti 2025): in my framework, economies of scale and scope emerge from posterior-variance reduction and the House Party Effect, without relying on demand-side feedback or exogenous complementarities.

The applications connect these statistical forces to the IO literature on digital markets, data pools, exclusive contracts, and scaling in machine learning. De Corniere and Taylor (2025) show that the pro- or anti-competitive effect of more data depends on the supply of data-driven services rather than demand, while Cornière and Taylor (2024) develop a theory of harm for data-driven mergers based on cross-market effects. Both treat data as an undifferentiated input; my analysis shows how their insights extend when "data" is unpacked into observations and distinct covariates. The pooling application relates to the patent-pool and broker literatures: Lerner and Tirole (2004) study the private and social value of commercializing complementary patents in pools, and Gu, Madio, and Reggiani (2021) analyze broker pools when datasets are substitutes. By endogenizing the statistical substitutability or complementarity of datasets, I microfound welfare implications that complement Nocke, Peitz, and Stahl (2007) on vertical integration and platform fragmentation. The analysis of exclusive data licenses builds on classic models in which exclusivity serves either to extract rents or to deter entry by raising rivals' costs (Katz and Shapiro 1986; Aghion and Bolton 1987), adapt-

ing these insights to non-rival training datasets that are strict complements to proprietary covariates.

2 Setup

An agent (she) seeks to predict an outcome for a target individual (he). To do so, she can collect (i) training data, consisting of outcomes and covariates for a sample of individuals, to learn the relationship between outcomes and covariates; and (ii) covariates of the target individual, to apply this relationship and form a prediction. The agent chooses which covariates to observe, trading off prediction accuracy and data-collection costs.

2.1 Data-Generating Process

An agent predicts a real-valued target variable y_0 for a target individual indexed by 0 and drawn from a countably infinite population indexed by $\mathbb{N}_0$. We consider a sequence of data-generating processes (DGPs) indexed by $Z \in \mathbb{N}_+$, where Z is the number of potential covariates. We first develop the model for a fixed, finite Z, and then study the implications as $Z \to \infty$. Fix Z. For each individual $i \in \mathbb{N}_0$, the target variable satisfies

$$y_i = \sum_{k=1}^{Z} \beta_k \, x_{ik} + \eta_i,$$

where $x_{ik} \in \mathbb{R}$ is covariate k of individual i, β_k is an unknown coefficient, and η_i is an idiosyncratic error term.

Coefficients. The coefficients are i.i.d. and independent of the covariates and errors:

$$\beta_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \tau^2\right),$$

where $\tau^2 \in [0, \infty)$ reflects prior uncertainty about the coefficients.

Covariates. Covariates are mutually independent and i.i.d. across individuals:

$$x_{ik} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{\sigma_x^2}{Z}\right).$$

The scaling by $1/Z$ keeps the total covariate variance $\sum_{k=1}^{Z} \mathrm{Var}(x_{ik})$ equal to σ_x^2, independently of Z. The parameter $\sigma_x^2 \geq 0$ reflects the overall signal in covariates.

Idiosyncratic Error. The idiosyncratic errors are i.i.d. across individuals and independent of coefficients and covariates:

$$\eta_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2 \tau^2),$$

where $\sigma_\eta^2 \geq 0$ reflects the importance of the *idiosyncratic noise* (e.g., measurement error or non-linearity in the DGP).
Under these assumptions, for all $i \in \mathbb{N}$, it holds that $\mathrm{Var}[y_i] = \tau^2(\sigma_x^2 + \sigma_\eta^2)$, which is finite and does not depend on Z. The linear specification should be interpreted as a reduced-form representation of the predictive component of a possibly nonlinear data-generating process. More generally, any square-integrable outcome admits a linear projection onto the covariates, and in high dimension with many weak predictors, smooth nonlinear functions are well approximated by their first-order expansion. Appendix **??** provides a brief justification.

Notation. For Z fixed, let $\boldsymbol{x}_i \equiv (x_{ik})_{k=1}^Z \in \mathbb{R}^Z$ denote the covariate vector of individual i, and let $\boldsymbol{\beta} \equiv (\beta_k)_{k=1}^Z \in \mathbb{R}^Z$ denote the coefficient vector. For any vector $\boldsymbol{v} \in \mathbb{R}^Z$ and index set $\mathcal{K} \subseteq \{1, \dots, Z\}$, let $\boldsymbol{v}_\mathcal{K} \equiv (v_k)_{k \in \mathcal{K}}$ denote the corresponding subvector.

2.2 Data and Predictor

The agent chooses three objects which determine the characteristics of the observed dataset:

- a ***sample size***, $N \in \mathbb{N}$, of individuals on which to observe the target variable and resulting in a ***target vector*** $\boldsymbol{y}_N \in \mathbb{R}^N$;
- a ***set of training covariates***, $\mathcal{T} \subseteq \{1, ..., Z\}$, observed on the N individuals in the sample and resulting in a ***design matrix*** $\boldsymbol{X}_{N,\mathcal{T}} \in \mathbb{R}^{N \times |\mathcal{T}|}$; and
- a ***set of prediction covariates***, $\mathcal{P} \subseteq \{1, ..., Z\}$, observed on the target individual 0 and resulting in an ***input vector*** $\boldsymbol{x}_{0,\mathcal{P}} \in \mathbb{R}^{|\mathcal{P}|}$.

The agent's data-collection strategy is summarized by the following object.

Definition 1. A ***dataset design*** is a triple

$$\mathcal{D} \equiv (N, \mathcal{T}, \mathcal{P}),$$

where N is the sample size, $\mathcal{T}$ is the set of training covariates and $\mathcal{P}$ is the set of prediction covariates.

Given a design $\mathcal{D}$, the ***training data*** consist of the pair

$$(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}),$$

where the i-th row of $\boldsymbol{X}_{N,\mathcal{T}}$ collects the covariates $(x_{ik})_{k \in \mathcal{T}}$ for individual i, and the i-th entry of $\boldsymbol{y}^N$ equals y_i. The ***dataset*** further includes the prediction covariates of the target individual:

$$\boldsymbol{D} \equiv ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}}).$$

Under design $\mathcal{D}$, the random dataset $\boldsymbol{D}$ is distributed according to the prior described in Section 2.1, which I denote by $P_{\mathcal{D}}$, i.e. $\boldsymbol{D} \sim P_{\mathcal{D}}$. We denote the *empty design* by $\varnothing$.

Predictor. Predictions depend on the realized dataset. A ***predictor*** is a measurable map $\hat{y} : \mathbb{R}^{N\times(1+|\mathcal{T}|)} \times \mathbb{R}^{|\mathcal{P}|} \to \mathbb{R}$ that assigns a prediction to each realized dataset:

$$\boldsymbol{D} \mapsto \hat{y}(\boldsymbol{D}).$$

In other words, under any fixed design $\mathcal{D}$, $\hat{y}$ is a random variable induced by the randomness in $\boldsymbol{D}$, and the predictor specifies how the agent uses the observed data to form a prediction.

2.3 Agent's Problem

Agent's Payoff The agent makes a point prediction $\hat{y} \in \mathbb{R}$ of the target outcome y. Her realized prediction error is $y - \hat{y}$, and she incurs quadratic loss

$$\ell(y, \hat{y}) = (y - \hat{y})^2 .$$

Therefore, the agent is penalized proportionally more for larger prediction errors. Because of its tractability, quadratic loss is a standard assumption in econometrics and machine learning. Furthermore, it has desirable theoretical properties discussed in Brier (1950) and axiomatized in Selten (1998).[6]

The agent minimizes her expected loss based on her knowledge of the data-generating process at the time of prediction. To increase her knowledge, she chooses a costly dataset design and observes the resulting dataset. For simplicity, I assume the total cost of a design depends only on the sample size and the number of training and prediction covariates, and is additively separable:

$$C_Z(\mathcal{D}) \equiv C_N\left(\frac{N}{Z}\right) + C_{\mathcal{T}}\left(\frac{|\mathcal{T}|}{Z}\right) + C_{\mathcal{P}}\left(\frac{|\mathcal{P}|}{Z}\right),$$

where $C_N(\cdot), C_{\mathcal{T}}(\cdot), C_{\mathcal{P}}(\cdot)$ are continuous, strictly increasing and positive. The cost may arise from the data collection cost, the data storage cost or the computation cost of training the model. I choose this simplistic form to impose minimal structure on results, making the qualitative predictions of the model depend exclusively on the improvements in prediction accuracy stemming from data collection (which captures the demand for data) rather than from the functional form of the cost (which reflects the supply of data).

I assume that the choice of a predictor is costless.

[6]Namely, among all incentive-compatible scoring rules—that is, rules for which truthful probabilistic forecasts maximize expected scores—the quadratic rule is uniquely characterized (up to a positive linear transformation) by three axioms: symmetry, elongation invariance, and neutrality.

Timing The agent's objective is to minimize the expected loss and the design cost. The predictor is chosen ex ante as a mapping from datasets to predictions. Therefore, the model unfolds in two steps:

1. ***Design Problem***: the agent chooses a design $\mathcal{D}$ and pays the associated cost;
2. ***Prediction Problem***: the agent chooses a predictor $\hat{y}$, nature draws a dataset $\boldsymbol{D}$ and an outcome y and losses occur.

As usual, I solve the model backwards. In the following section, I provide an overview of the problem the agent faces. I defer the solution to Sections 3 and 4.2.

2.3.1 Prediction Problem

The agent chooses the predictor so that conditional on the realization of a dataset $\boldsymbol{D}$, she minimizes her posterior risk, defined as the expected loss conditional on $\boldsymbol{D}$:

$$\mathbb{E}\left[\ell(y, \hat{y}(\boldsymbol{D})) \mid \boldsymbol{D}\right].$$

Therefore, the agent chooses the ***optimal predictor***

$$\hat{y}^*(\boldsymbol{D}) \equiv \arg\min_{\hat{y}} \mathbb{E}\left[\ell(y, \hat{y}(\boldsymbol{D})) \mid \boldsymbol{D}\right].$$

For quadratic loss, $\hat{y}^*$ exists and is unique up to realizations of $\boldsymbol{D}$ with probability 0. Having characterized the optimal predictor, I can define the ***value of a dataset*** $\boldsymbol{D}$ as

$$v(\boldsymbol{D}) \equiv \underbrace{\mathbb{E}\left[\ell(y, \hat{y}^*(\varnothing))\right]}_{\text{minimum prior risk}} - \underbrace{\mathbb{E}\left[\ell(y, \hat{y}^*(\boldsymbol{D})) \mid \boldsymbol{D}\right]}_{\text{minimum posterior risk}}.$$

The value of a dataset $\boldsymbol{D}$ is the reduction of risk that the agent achieves by observing it and using it optimally to predict $\hat{y}^*(\boldsymbol{D})$, relative to the outside option of not collecting any data and making the ex-ante optimal prediction $\hat{y}^*(\varnothing)$.

2.3.2 Dataset Design Problem

The value of a dataset design is the ex-ante expected value of the dataset it induces.

Definition 2. The ***value of a design*** $\mathcal{D}$ is

$$V(\mathcal{D}) \equiv \mathbb{E}_{\boldsymbol{D} \sim P_{\mathcal{D}}}\left[v(\boldsymbol{D})\right].$$

The value of a design $\mathcal{D}$ is the maximal increase in ex-ante expected payoff that the agent can achieve when using the optimal predictor $\hat{y}^*(\boldsymbol{D})$, given that design.

Maximizing the payoff minus data collection costs amounts to maximizing the value of a design minus its costs. Therefore, the agent solves

$$\max_{\boldsymbol{D}} \{V(\boldsymbol{D}) - C(\boldsymbol{D})\},$$

which, I show, always has a solution.

3 Low-Dimensional Data-Generating Process

In this section I characterize the optimal predictor and the value of data in a DGP with a fixed, finite number of covariates $Z \in \mathbb{N}_+$. The agent chooses how to use the input vector $\boldsymbol{x}_{\mathcal{P}}$ of the target individual and the training data $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ to form a prediction.

The optimal predictor separates (i) *learning across individuals* from the training sample, which updates beliefs about $\boldsymbol{\beta}$, and (ii) *using within-individual information* $\boldsymbol{x}_{\mathcal{P}}$, which maps posterior beliefs into a prediction.

Under quadratic loss, the optimal predictor is the posterior mean of the target variable, a standard result in Bayesian decision theory.

Proposition 1 (Optimal predictor)**.** *The optimal predictor is*

$$\hat{y}^*(\boldsymbol{D}) = \mathbb{E}\left[y \mid \boldsymbol{D}\right] = \sum_{k \in \mathcal{P} \cap \mathcal{T}} x_{0,k} \mathbb{E}\left[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right].$$

The identity $\hat{y}^*(\boldsymbol{D}) = \mathbb{E}[y \mid \boldsymbol{D}]$ follows from the standard optimal prediction rule under squared loss. Because the target variables are conditionally independent across individuals given $\boldsymbol{\beta}$, the conditional expectation $\mathbb{E}[y \mid \boldsymbol{D}]$ depends on the training data $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ only through the posterior distribution of the coefficients.

Furthermore, covariates and coefficients are mutually independent across indices. Hence, for any coefficient $k \in \mathcal{P} \setminus \mathcal{T}$ that does not appear in the training data, the likelihood of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ does not depend on β_k, implying that its posterior equals its prior. Since the prior mean is zero,

$$\mathbb{E}\left[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right] = 0, \quad \forall k \in \mathcal{P} \setminus \mathcal{T}.$$

Therefore, covariates that are observed for prediction but not included in the training set do not affect the optimal predictor.

Given quadratic loss, Proposition 1 implies that the value of a dataset equals the reduction in posterior uncertainty about the target variable.

Proposition 2 (Value of a dataset)**.** *The value of a dataset* $\boldsymbol{D}$ *is*

$$v(\boldsymbol{D}) = \operatorname{Var}[y] - \operatorname{Var}[y \mid \boldsymbol{D}] = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^{\top} \left(\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}] - \operatorname{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})]\right) \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

The value of a dataset is driven by how much it reduces uncertainty about the parameters of the observed covariates. The bracketed matrix captures the posterior reduction in the covariance of $\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}$ induced by observing $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$. The quadratic form $\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top(\cdot)\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}$ then maps this reduction in coefficient uncertainty into a reduction in uncertainty about the target outcome for the specific individual under consideration.
Intuitively, a dataset is most valuable when the input vector $\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}$ loads heavily on covariates whose coefficients are learned most precisely from the training data, and its value vanishes either when $\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}} = \mathbf{0}$ or when the posterior coincides with the prior.
Proposition 4 has an immediate implication for optimal data design. As observing any covariate for prediction entails a strictly positive cost, an optimal design never includes a prediction covariate that is absent from the training set.

Corollary 1 (No useless prediction covariates)**.** *In any optimal design it must hold that $\mathcal{P} \subseteq \mathcal{T}$.*

Remark 1. Given that I have assumed index-wise symmetry (covariates and coefficients are i.i.d. across k and costs depend only on set cardinalities), the triplet $(N, |\mathcal{T}|, |\mathcal{P}|)$ characterizes a dataset design up to a relabeling of covariate indices. That is, the identities of the observed covariates are irrelevant; only the sample size, the number of covariates in the training and prediction set matter.

Without loss of generality, I therefore assume that $\mathcal{T} = \{1, \dots, |\mathcal{T}|\}$ and $\mathcal{P} = \{1, \dots, |\mathcal{P}|\}$, with $|\mathcal{T}| \geq |\mathcal{P}|$ by Corollary 1.

4 High-Dimensional Data-Generating Process

Given Z and conditional on $\mathcal{T}(Z)$, for any individual i in the training sample,

$$y_i^{(Z)} = \sum_{k=1}^{|\mathcal{T}(Z)|} \beta_k^{(Z)} x_{ik}^{(Z)} + \varepsilon_i^{(Z)}, \qquad \varepsilon_i^{(Z)} \equiv \sum_{k=|\mathcal{T}(Z)|+1}^{Z} \beta_k^{(Z)} x_{ik}^{(Z)} + \eta_i^{(Z)}.$$

The *regression residual*, $\varepsilon_i^{(Z)}$, consists of the contribution of omitted covariates, $\sum_{k=|\mathcal{T}(Z)|+1}^{Z} \beta_k^{(Z)} x_{ik}^{(Z)}$, and the idiosyncratic error, $\eta_i^{(Z)}$. Its distribution is generally complicated, as it is a sum of products of independent normal variables. However, I show that, in the limit as $Z \to \infty$, the distribution of the residual becomes tractable.
To do so, I define a *sequence of dataset designs* as a collection $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ and make the following asymptotic assumption.

Assumption 1 (Proportional-growth design)**.** *We restrict attention to sequences $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ for which there exist constants $(n, t, p) \in (0, \infty) \times [0, 1] \times [0, 1]$ such that, as $Z \to \infty$,*

$$\frac{N(Z)}{Z} \to n, \qquad \frac{|\mathcal{T}(Z)|}{Z} \to t, \qquad \frac{|\mathcal{P}(Z)|}{Z} \to p.$$

We call (n, t, p) *the* (asymptotic) design dimensions *of the sequence, with* n *the* (asymptotic) training scale, t *the* (asymptotic) training scope, *and* p *the* (asymptotic) prediction scope.

Assumption 1 imposes a *proportional-growth* (or *linear-scaling*) regime: as the number of potential covariates Z grows, the sample size $N(Z)$ and the sizes of the training and prediction scopes $|\mathcal{T}(Z)|$ and $|\mathcal{P}(Z)|$ all grow at rate Z, with stable limiting shares (n, t, p). This is the standard asymptotic regime used in random matrix theory (see, e.g., Marčenko and Pastur (1967), Bai and Silverstein (2009), and Hastie et al. (2020)), as it delivers tractable limits for the posterior variances that drive prediction risk.[7]

Moreover, Assumption 1 reflects the abundance of data that characterizes modern machine learning environments, where both the number of observations and the number of covariates are typically large.

The following result characterizes the asymptotic distribution of the regression residual as $Z \to \infty$.

Lemma 1. *For any sequence of dataset designs* $\{\mathcal{D}(Z)\}_{Z \in \mathbb{N}_+}$ *satisfying Assumption 1,*

$$(\varepsilon_1^{(Z)}, \dots, \varepsilon_N^{(Z)})^\top \underset{Z \to \infty}{\overset{d}{\to}} \mathcal{N}(\mathbf{0}_N, \sigma_\varepsilon^2(t)\ \tau^2 \cdot \boldsymbol{I}_N),$$

where the (relative) residual noise $\sigma_\varepsilon^2(t)$ *satisfies*

$$\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2.$$

When Z is large and the training scope covers a fraction t of covariates, the residual contains the contribution of a large number of omitted covariates (a fraction $1-t$ of them). Omitted covariates are mutually independent and each of them has a small impact on the target variable because they are scaled so that total signal stays fixed as Z grows. A standard averaging/central-limit logic then implies that the sum of many small, independent omitted-covariate contributions behaves approximately like a Gaussian noise term. Its variance $\sigma_\varepsilon^2(t)$ is increasing in the share of omitted covariates $1-t$: omitting more covariates (smaller t) leaves more unmodeled signal in the residual and thus raises its dispersion, while expanding the training scope (larger t) absorbs more of that signal and reduces residual uncertainty. The parameter σ_x^2 governs the strength of this effect by scaling the overall amount of covariate-driven variation that enters in the residual. The independence across individuals implies that residuals are (asymptotically) independent across observations.

Lemma 1 implies that, asymptotically, the training data satisfies a standard Gaussian linear regression with homoskedastic errors. Consequently, the problem of estimating $\boldsymbol{\beta}$ reduces to a textbook Bayesian linear regression problem (e.g., DeGroot (1970) and Berger (1990)), and the posterior distribution of $\boldsymbol{\beta}$ can be characterized using standard results.

[7] I depart from the standard regime in that I track three asymptotic dimensions, reflecting the fact that the agent controls three distinct margins: the number of training observations $N(Z)$, the number of training covariates $|\mathcal{T}(Z)|$, and the number of prediction covariates $|\mathcal{P}(Z)|$.

4.1 The Value of a Dataset

Proposition 3 (Optimal Predictor)**.** *The optimal predictor is*

$$\hat{y}^*(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \underbrace{\left(\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\cdot \boldsymbol{I}_{|\mathcal{T}|}\right)^{-1} \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{y}_N\right)_{\mathcal{P}\cap\mathcal{T}}}_{=\mathbb{E}\left[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}|(\boldsymbol{y}_N,\boldsymbol{X}_{N,\mathcal{T}})\right]}.$$

The posterior mean of the coefficients is a shrinkage of the MLE estimator

$$\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}}\right)^{-1} \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{y}_N$$

towards the prior mean $\boldsymbol{0}_{|\mathcal{T}|}$. The strength of this shrinkage is governed by the residual noise $\sigma_\varepsilon^2(t)$. Intuitively, if $\sigma_\varepsilon^2(t) \approx 0$, the coefficients $\boldsymbol{\beta}_\mathcal{T}$ are estimated precisely, so the posterior mean coincides with the maximum-likelihood estimator. As $\sigma_\varepsilon^2(t)$ increases, the agent becomes more cautious in using the training data, and the posterior mean shrinks more strongly towards the prior mean.

The residual noise $\sigma_\varepsilon^2(t)$ is equivalent to the misspecification parameter of Strzalecki (2024). In contrast to his model, where the misspecification parameter is exogenous, my framework endogenizes $\sigma_\varepsilon^2(t)$ by letting it depend on the training scope of the dataset design chosen by the agent.

Given the loss is quadratic, Proposition 1 implies that the value of a dataset is the reduction in the variance of the target variable brought forth by its observation, so I can state the following result.

Proposition 4. *The value of a dataset $\boldsymbol{D}$ is*

$$v(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \underbrace{\left(\tau^2 \cdot \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}} \left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2 \cdot \boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}\right)_{\mathcal{P}\cap\mathcal{T}}}_{=\ \mathrm{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}] - \mathrm{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}|(\boldsymbol{y}_N,\boldsymbol{X}_{N,\mathcal{T}})]} \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

The value of a given dataset $\boldsymbol{D}$ is decreasing and convex in the residual noise $\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2$. Therefore, the marginal value of reducing residual noise is larger when σ_ε^2 is already low. Since $\sigma_\varepsilon^2(t)$ decreases linearly in t at rate σ_x^2, expanding the training scope is especially valuable when t is already large and when σ_x^2 is high. This feature is a key driver of increasing returns to the scope of training covariates, which I explore in the next section.

4.2 Training Information

To study the optimal design choice, I first characterize how informative a design is about the coefficients, and then translate this into the value of a covariate set and the marginal value of individual covariates.

4.2.1 Definition and Characterization of Training Information

We first study how informative the training data $(\boldsymbol{y}, \boldsymbol{X}_{\mathcal{T}})$ is about the coefficient vector $\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}$.

Definition 3. Fix a proportional-growth sequence $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ with $N(Z)/Z \to n$ and $|\mathcal{T}(Z)|/Z \to t$. The ***(asymptotic) training information*** is

$$\mathcal{I}(n,t) \equiv 1 - \lim_{Z\to\infty} \frac{1}{\tau^2\,|\mathcal{T}(Z)|}\,\mathbb{E}_{D\sim P_{D(Z)}}\left[\operatorname{tr}\Big(\operatorname{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(Z)} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}(Z)}\big)\Big)\right],$$

provided the limit exists.

The object $\mathcal{I}(n,t)$ measures how much the training data $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ shrinks uncertainty about the coefficients, per trained covariate, on average, in the proportional-growth regime. Because the prior covariance of each trained coefficient is τ^2, $\mathcal{I}(n,t)$ is the average fraction of prior variance learned by the data. Thus $\mathcal{I}(n,t) = 0$ means training is essentially uninformative (posterior variance is close to prior variance), while $\mathcal{I}(n,t) \approx 1$ means training is highly informative (posterior variance is close to zero).
The following result characterizes $\mathcal{I}(n,t)$ using the spectral properties of high dimensional matrices and the Marchenko–Pastur distribution.

Proposition 5 (Closed Form Information)**.** *Under Assumption 1, the training information satisfies*

$$\mathcal{I}(n,t) = \frac{n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1}{2t}\left(1 - \sqrt{1 - \frac{4nt}{\left(n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1\right)^2}}\right),$$

Equivalently, $\mathcal{I}(n,t)$ is the unique solution in $[0,1]$ to

$$\mathcal{I} = \frac{1}{1 + \dfrac{\overbrace{\sigma_\varepsilon^2(t) + t\sigma_x^2(1-\mathcal{I})}^{\text{Noise}}}{\underbrace{n\sigma_x^2}_{\text{Signal}}}},$$

where $\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2$. This implicit formulation highlights that $\mathcal{I}(n,t)$ is the share of total variation attributable to $n\sigma_x^2$ rather than the effective noise which is the sum of two terms:

- The *residual noise* (idiosyncratic noise plus omitted covariates)

  $$\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2,$$

  which pollutes the coefficient estimates, reducing $\mathcal{I}$;

- An *estimation noise*

$$t\sigma_x^2(1-\mathcal{I})$$

that is the uncertainty on the coefficients which remains after learning. The agent estimates a fraction t of the coefficients. Each estimated coefficient contributes to the variance of the target variable with weight σ_x^2, and retains a posterior variance equal to a fraction $(1-\mathcal{I})$ of its prior variance. The aggregate uncertainty that remains in these coefficients therefore generates predictive noise proportional to $t\sigma_x^2(1-\mathcal{I})$.

Therefore, the dependence of information on the training scale and scope can be analyzed by taking derivatives. The following result characterizes the impact of increasing the number of observations (the scale) on the information.

Corollary 2 (Scale and Information)**.** *For any $n \geq 0$ and $t \in [0,1]$,*

$$\frac{\partial}{\partial n}\mathcal{I}(n,t) > 0 \quad \textit{and} \quad \frac{\partial^2}{\partial n^2}\mathcal{I}(n,t) < 0.$$

Training information is increasing and concave in the training scale n: collecting more observations improves learning, but at diminishing marginal returns. The intuition is that, holding the training covariate set fixed, additional observations are i.i.d. draws, so the marginal gain from enlarging the sample shrinks as n grows.

The following result characterizes the impact of increasing the number of observed covariates (the scope) on the information.

Corollary 3 (Scope and Information)**.** *For any $t \in [0,1]$,*

$$\frac{\partial}{\partial t}\mathcal{I}(n,t) > 0 \quad \textit{and} \quad \frac{\partial^2}{\partial t^2}\mathcal{I}(n,t) > 0.$$

Increasing t impacts the information in two opposite ways: (i) it reduces *residual noise*, as there are fewer omitted covariates (positive effect on information); (ii) it increases *estimation noise*, as more coefficients must be estimated from a sample of a given size, resulting in a loss of degrees of freedom (negative effect of information). To study which effect dominates, I simplify previous equation to

$$\mathcal{I} = \frac{1}{1+\frac{1+\frac{\sigma_\eta^2}{\sigma_x^2}-t\mathcal{I}}{n}}.$$

Inspection of this equation reveals that the residual noise reduction effect always dominates so $\mathcal{I}$ is increasing in t: the reason is that the marginal reduction in the residual noise from an increase in t is σ_x^2; the marginal increase in estimation noise per additional coefficient is $\sigma_x^2(1-\mathcal{I})$, but since $\mathcal{I} \in [0,1]$, the former term dominates, and the information is increasing in t.

The equation also clarifies the source of the convexity of $\mathcal{I}(n,t)$ with respect to t: $\mathcal{I}$ enters the effective noise term through $t\sigma_x^2(1-\mathcal{I})$. An increase in t reduces residual noise directly,

which raises $\mathcal{I}$; a higher $\mathcal{I}$ in turn lowers estimation noise, further increasing $\mathcal{I}$. Because this amplification operates through all trained coefficients, and the number of trained coefficients grows with t, the marginal gain from expanding scope rises with t, generating convexity.

Corollary 4 (Complements/substitutes)**.** *For each $t \in [0,1]$, there is a decreasing threshold $\hat{n}(t) \in [0,\infty]$ such that*

$$\frac{\partial^2}{\partial n \partial t}\mathcal{I}(n,t) \gtreqless 0 \quad \textit{iff} \quad n \lesseqgtr \hat{n}(t),$$

where $\hat{n}(t) \in (\tilde{n},\infty)$ is unique and decreasing in t.

The result can be summarized in the following figure.

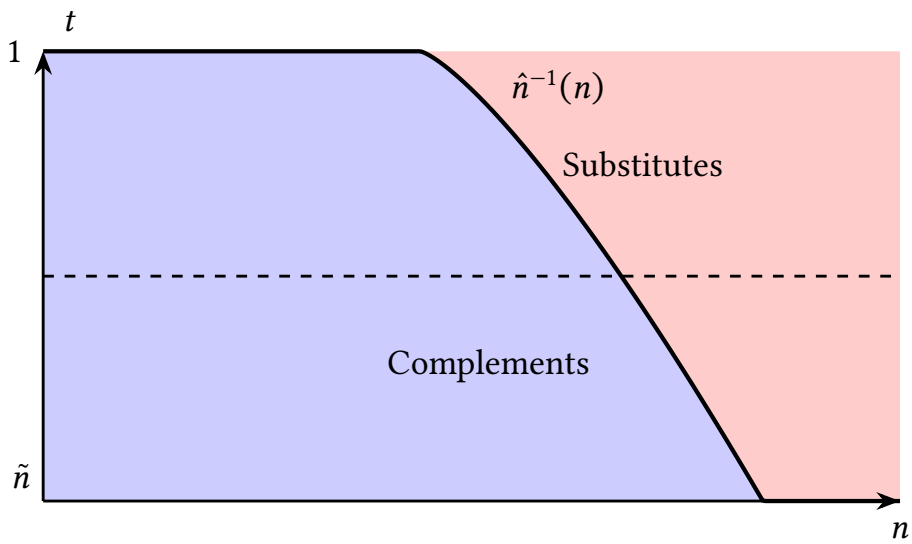

Figure 1: Below the decreasing threshold $t = \hat{n}^{-1}(n)$, observations and covariates are complements ($\partial^2\mathcal{I}/\partial n \partial t \geq 0$); above it they are substitutes.

To gain intuition on why the modularity between training scope and scale changes, consider that increasing the scope affects the relationship between information and the scale in two opposite ways

- By increasing the number of coefficients that must be estimated, a larger scope reduces the degrees of freedom, thereby increasing the positive effect of a larger sample on the information. This force drives *complementarity* between scale and scope;
- By decreasing the number of unobserved covariates a larger scope decreases the residual noise, thereby decreasing the positive effect of a larger sample on the information, which hinges on its role to "average out" the noise. This force drives *substitutability* between scale and scope.

Below the threshold $\hat{n}(t)$, there are few degrees of freedom, so scope and scale are complements. Above the threshold the opposite holds, so scope and scale are substitutes.

4.3 The Value of a Design

Definition 4 (Asymptotic value and cost)**.** Fix a sequence of designs $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ satisfying Assumption 1 and with asymptotic design dimensions (n, t, p). The ***asymptotic design value*** is

$$\mathcal{V}(n, t, p) \equiv \lim_{Z\to\infty} V(\mathcal{D}(Z)),$$

provided the limit exists. The ***asymptotic cost*** is

$$\mathcal{C}(n, t, p) \equiv \lim_{Z\to\infty} C_Z(\mathcal{D}(Z)).$$

Whenever $\mathcal{V}(n, t, p)$ is well-defined, the agent's asymptotic design problem is

$$\max_{n>0,\ 0\le p\le t\le 1} \Big\{ \mathcal{V}(n, t, p) - \mathcal{C}(n, t, p) \Big\}.$$

If $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ is a sequence of designs satisfying Assumption 1 with asymptotic design dimensions (n, t, p), then

$$\mathcal{C}(n, t, p) = C_N(n) + C_{\mathcal{T}}(t) + C_{\mathcal{P}}(p).$$

I now characterize the asymptotic value of a design.

Theorem 1 (Asymptotic design value)**.** *Let $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ be a sequence of designs satisfying Assumption 1 with asymptotic design dimensions (n, t, p), with $n \ge 0$ and $1 \ge t \ge p \ge 0$. Then the asymptotic value exists and satisfies*

$$\mathcal{V}(n, t, p) = \underbrace{\sigma_x^2\, p}_{\textit{usable target signal}} \times \underbrace{\tau^2\, \mathcal{I}(n, t)}_{\textit{learning per trained coefficient}}.$$

where $\mathcal{I}(n, t)$ is the relative training information defined in Definition 3.

The asymptotic value of a design is the product of two components:

- *Usable target signal.* The number of prediction covariates is asymptotically Zp, and since each covariate has variance σ_x^2/Z, the total usable covariate signal for the target individual is $\sigma_x^2 p$, as Corollary 1 implies I can restrict attention to $0 \le p \le t \le 1$.
- *Learning per trained coefficient.* The training scope and scale determine how precisely the coefficients on trained covariates are estimated, and enter through the training information $\tau^2\, \mathcal{I}(n, t)$.

4.4 Returns to Design Dimensions

I study the returns to dataset design dimensions by differentiating the closed form of Theorem 1.

Corollary 5 (Returns to Scale). *For any* $n \geq 0$ *and* $0 \leq p \leq t \leq 1$,

$$\frac{\partial}{\partial n}\mathcal{V}(n,t,p) = \sigma_x^2\, p\tau^2 \frac{\partial}{\partial n}\mathcal{I}(n,t) > 0 \quad \textit{and} \quad \frac{\partial^2}{\partial n^2}\mathcal{V}(n,t,p) = \sigma_x^2\, p\tau^2 \frac{\partial^2}{\partial n^2}\mathcal{I}(n,t) < 0.$$

The value is increasing and concave in n because it is linear in $\mathcal{I}(n,t)$ and depends on n only through the latter, which is increasing and concave in n. This is consistent with the empirical and conceptual evidence that additional data points are most valuable at low scale and become less valuable as scale grows (e.g., Bajari et al. (2019), Goldfarb and Tucker (2011), Schaefer and Sapi (2023), and Lee and Wright (2023)).

Corollary 6 (Returns to Prediction Scope). *For any* $n \geq 0$ *and* $0 \leq p \leq t \leq 1$,

$$\frac{\partial}{\partial p}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2\, \mathcal{I}(n,t)\mathbf{1}\left(\{p < t\}\right) \geq 0 \quad \textit{and} \quad \frac{\partial^2}{\partial p^2}\mathcal{V}(n,t,p) = 0.$$

The value is weakly increasing in p. It is strictly increasing in p if $p < t$, because in this case the agent can add prediction covariates the coefficients of which it has information on. The returns to scale are constant: since covariates are i.i.d. each covariate contributes equally to the overall variance of the target variable.

Corollary 7 (Returns to Training Scope). *For any* $n \geq 0$ *and* $0 \leq p \leq t \leq 1$,

$$\frac{\partial}{\partial t}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2 p\frac{\partial}{\partial t}\mathcal{I}(n,t) > 0 \quad \textit{and} \quad \frac{\partial^2}{\partial t^2}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2 p\frac{\partial^2}{\partial t^2}\mathcal{I}(n,t) > 0.$$

Increasing the training scope t increases the information per coefficient: each additional covariate reduces residual noise enough that learning about the whole set of trained coefficients improves. Moreover, this improvement accelerates as t grows (increasing returns): once the model is already rich, adding another covariate helps inference on many coefficients at once. These economies of scope are consistent with the empirical findings in Carballa-Smichowski et al. (2025b).

Corollary 8 (Complements/Substitutes). *For any* $n \geq 0$ *and* $0 \leq p \leq t \leq 1$,

$$\frac{\partial^2}{\partial n \partial p}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2\frac{\partial}{\partial n}\mathcal{I}(n,t)\mathbf{1}\left(\{p < t\}\right) \geq 0,$$

$$\frac{\partial^2}{\partial t \partial p}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2\frac{\partial}{\partial t}\mathcal{I}(n,t)\mathbf{1}\left(\{p < t\}\right) \geq 0,$$

and

$$\frac{\partial^2}{\partial n \partial t}\mathcal{V}(n,t,p) = \tau^2\sigma_x^2 p\frac{\partial^2}{\partial n \partial t}\mathcal{I}(n,t) \lesseqgtr 0 \iff n \lesseqgtr \hat{n}(t),$$

where $\frac{\partial^2}{\partial n \partial t}\mathcal{I}(\hat{n}(t),t) = 0$.

Scale and Prediction Scope. Observations and prediction covariates are complements. The reason is that a larger scale implies more information on the covariates' correlation with the target variable, so it increases their value. This finding confirms the empirical results in Schaefer and Sapi (2023) and Lee and Wright (2023), which show that the marginal value of an observation is increasing in the number of covariates (and vice versa).

Training and Prediction Scope. Training and prediction covariates are complements because observing additional training covariates (larger t) increases the information on the coefficients of the prediction covariates.

Training Scale and Scope. If the scale and training scope are sufficiently large, observations and training covariates can be substitutes. The reason is that both training scope and scale concur to improve the information on the correlation of the covariates and the target variable, and when the training dataset is large enough, they are substitutes. Furthermore, this result implies that the complementarity between covariates and observations empirically documented in Schaefer and Sapi (2023) and Lee and Wright (2023) may not apply to training covariates when there are many observations and training covariates.

4.5 Optimal Dataset Design

The agent chooses dataset scale and scope (n, t, p) to maximize net surplus

$$\max_{n\geq 0,\ 0\leq p\leq t\leq 1} \mathcal{V}(n,t,p) - \mathcal{C}(n,t,p),$$

where

$$\mathcal{V}(n,t,p) = \tau^2\sigma_x^2\, p\, \mathcal{I}(n,t), \qquad \mathcal{C}(n,t,p) = C_N(n) + C_{\mathcal{T}}(t) + C_{\mathcal{P}}(p),$$

and $C_N, C_{\mathcal{T}}, C_{\mathcal{P}}$ are positive, strictly increasing, continuous and convex, with $\lim_{n\to\infty} C_N(n) = \infty$. Under these restrictions on the cost function $\mathcal{C}(n,t,p)$, a non-trivial optimal design always exists and is characterized in the following proposition.

Proposition 6 (Characterization of the optimal design)**.** *Let (n^*, t^*, p^*) be a local optimum of $\Pi(n,t,p)$.*

1. *If $n^* > 0$ and $0 < p^* < t^* < 1$, then*

$$C_N'(n^*) = \tau^2\sigma_x^2\, p^*\, \mathcal{I}_n(n^*, t^*), \tag{1}$$

$$C_{\mathcal{P}}'(p^*) = \tau^2\sigma_x^2\, \mathcal{I}(n^*, t^*), \tag{2}$$

$$C_{\mathcal{T}}'(t^*) = \tau^2\sigma_x^2\, p^*\, \mathcal{I}_t(n^*, t^*). \tag{3}$$

2. *If* $p^* = t^* \in (0,1)$ *and* $n^* > 0$, *then*

$$C_N'(n^*) = \tau^2 \sigma_x^2 \, t^* \, \mathcal{I}_n(n^*, t^*), \tag{4}$$

$$C_{\mathcal{T}}'(t^*) + C_{\mathcal{P}}'(t^*) = \tau^2 \sigma_x^2 \big(\mathcal{I}(n^*, t^*) + t^* \, \mathcal{I}_t(n^*, t^*) \big). \tag{5}$$

Second-order conditions for local optimality are verified in the proof.

The proposition characterizes the optimal choice of the three distinct margins of data design.

Training Scale. Increasing sample size raises learning through $\mathcal{I}(n,t)$ but at diminishing returns, so optimal scale equates marginal learning gains to the marginal cost $C_N'(n)$.

Prediction Scope. For $p < t$, prediction scope has constant and positive marginal value $\tau^2 \sigma_x^2 \mathcal{I}(n,t)$. For $p = t$, prediction scope has zero marginal value as using untrained covariates as prediction covariates brings no additional value. Prediction scope and scale are always complements, since $\partial_{pn} \mathcal{V} > 0$. Therefore increases in the cost of observations reduce the optimal prediction scope and vice versa.

Training Scope. Expanding training scope improves inference. It may therefore be optimal to push the training scope beyond the prediction scope, leading to interior optima with $p^* < t^*$. Models which are trained on many covariates and then deployed using only a subset of covariates are broadly used in practice and are referred to as "train-large, deploy-small" models. Whether the solution is interior or along the corner $p = t$, there are always increasing returns to scope, reflecting that additional covariates both strengthen learning and add usable signal. Furthermore, by Corollary 8, training scope and scale can be substitutes if the costs of t and n are low enough: then $n^* > \tilde{n}(t^*)$ and increases in the cost of observations *increase* the number of training covariates and vice versa.

5 Applications

This section applies the model in an observation-rich environment in which $n \gg t$, so that estimation noise is negligible. Accordingly, we work with the following approximation to the value of a dataset design.

Lemma 2 (Observation-rich approximation for value)**.** *Fix* $p \in (0,1]$ *and suppose that* $t/n \to 0$. *Then, with endogenous residual variance* $\sigma^2(t) = 1 - t(1 - \underline{\sigma}^2)$,

$$\bar{V}(n,t,p) = (1 - \underline{\sigma}^2) \frac{\min\{t,p\}}{1 + \sigma^2(t)/n} + o(1). \tag{6}$$

Throughout the applications, we restrict attention to balanced dataset designs in which the training and prediction covariates coincide. That is, we assume $\mathcal{T} = \mathcal{P} = \mathcal{K}$, and define

$$k \equiv \lim_{Z \to \infty} \frac{|\mathcal{K}(Z)|}{Z},$$

so that $t = p = k$. Under this restriction, the value of a dataset design is given by $\bar{V}(n, k)$.

5.1 Natural Monopsony and Data-driven Acquisitions

In this application we show how a "natural monopsony" in data can emerge: a firm with more covariates has a strictly increasing and convex advantage in acquiring additional covariates. This leads to inevitable acquisitions even when products, users, and markets do not overlap. Mergers may be anti-competitive even when the merging firms do not compete for users, attention, advertising, or any traditional IO market.

5.1.1 Setup

Demand. For each prediction seller there is a unit mass of captive prediction buyers.[8] Each buyer must choose an action $\hat{y} \in \mathbb{R}$, while the payoff-relevant state is $y \in \mathbb{R}$. Payoffs depend on the squared error $(\hat{y} - y)^2$. Buyers share a common prior on y as specified in Section 2. If they do not buy any prediction, they choose the prior mean and obtain expected utility

$$\bar{u} = -1,$$

because of the normalization $\mathrm{Var}(y) = 1$.
If a buyer purchases from seller i at price p_i, her *net* surplus relative to the outside option is

$$u(p_i, V_i) = V_i - p_i,$$

where $V_i \in [0, 1]$ is the value of the dataset design of seller i, which is equivalent to the buyers' willingness-to-pay for seller i's prediction: each buyer strictly prefers i whenever $p_i < V_i$. We summarize individual and competitive demand by a reduced-form function $D(p_i, V_i)$,which is increasing in V_i and decreasing in p_i.

Supply. There are two potential sellers: an incumbent I ("Big Tech") and an entrant E ("Fitbit"). Both observe the same sample size n. Seller $i \in \{I, E\}$ has access to a set $\mathcal{K}_i$ of training and prediction covariates, with disjoint covariate sets $\mathcal{K}_I \cap \mathcal{K}_E = \varnothing$. To obtain closed-form expressions, assume all covariates have identical informativeness and impose symmetry by

[8] I assume there is no downstream competition to abstract from the usual "killer acquisition" motive: acquisitions are solely motivated by the acquisition of data.

letting

$$|\mathcal{K}_I| = |\mathcal{K}_E| = k/2,$$

so that firms are symmetric.
The standalone value of seller i's design is $V_1 = V(n, \mathcal{K}_i)$.
The entrant E can either: (i) remain active and sell its own predictions, or (ii) accept a take-it-or-leave-it acquisition offer P from I. If I acquires E, it gains access to E's covariates. The merged firm operates with sample size n and combined covariate sets, so post-merger design value is

$$V_2 = V(n, \mathcal{K}_I \cup \mathcal{K}_E).$$

Profits. Let $D(p, V)$ denote the demand faced by a monopolist offering a prediction with design value V at price p. There are two possible cases

- *If I acquires E:* Only the merged seller operates. Setting price p_2, the incumbent's profit is

$$\Pi_I^{\text{acq}}(P, p_2) = p_2 D\big(p_2, V_2\big) - P,$$

  while the entrant receives the acquisition payment

$$\Pi_E^{\text{acq}}(P) = P.$$

- *If there is no acquisition:* Both firms are active. As demands are not related because each firm has a unit mass of captive customers, let $D_i(p_i, V_i)$ be seller i's demands. Profits are

$$\Pi_i^{\text{no}}(p_i) = p_i D_i(p_i, V_i).$$

Planner welfare. The planner's objective is

$$W = \Pi_I + \Pi_E + CS - \xi \times \mathbf{1}\big(I \text{ acquires } E\big),$$

where CS is customer surplus and $\xi \geq 0$ is the social cost of reduced entry. Intuitively, ξ reflects positive spillovers, e.g., new data, experimentation on alternative prediction tasks, technological diversity, knowledge spillovers.

Timing. The game unfolds in two stages:

1. **Acquisition stage.** The incumbent I offers P to acquire E. The entrant accepts or rejects.

2. **Pricing stage.** If the acquisition occurs, the merged firm chooses p_2. Otherwise, I and E simultaneously set (p_I, p_E).

Payoffs are realized at the end. The solution concept is subgame perfect Nash equilibrium, obtained by backward induction.

5.1.2 Buyer Purchase

Prediction Sale Buyers purchase the prediction if and only if $V \geq p$. The resulting demand function is

$$D(p, V) = \mathbf{1}(p \leq V).$$

The seller therefore sets

$$p^* = V,$$

sells to all buyers, and obtains profit equal to the value of its dataset design:

$$\Pi_i^{\text{no}} = V_i.$$

Intuitively, with homogeneous buyers and no frictions, the platform can fully extract the expected gains from improved prediction.

Acquisition For the entrant E, the standalone outside option is the expected gains from improved prediction when it operates separately from I, earning

$$\Pi_E^{\text{no}} = V_1.$$

To acquire E, it is sufficient that the incumbent I offer it this amount. When I acquires E, it combines covariates and gains an increase of value

$$V_2 - V_1$$

compared to its outside option V_1.

Proposition 7 (Harmful Acquisition)**.** *The incumbent I always acquires the entrant E.*

Intuitively, the economies of scope stemming from Corollary **??** implies that

$$V_2 - \sum_{i \in \{I,E\}} V_i \geq 0.$$

Combining disjoint datasets makes each covariate more informative. This is due to two things: first, as far as prediction covariates $\mathcal{K}_i$ can benefit from the covariate spillovers of the training covariates $\mathcal{K}_{-i}$ (and vice versa); second, due to the House Party Effect, pushing training covariates from $\mathcal{K}_i$ to $\mathcal{K}_I \cup \mathcal{K}_E$ yields supermodular gains. As the incumbent can fully exploit these complementarities, it will prevent entry by E even if demands are independent. This is

the natural-monopsony effect: data-driven economies of scope push data into the hands of a single firm.

5.1.3 Planner's Problem

With homogeneous buyers and full surplus extraction, customer surplus is zero in equilibrium. Social welfare therefore equals industry profit net of any cost of foregone entry and spillovers:

$$W = \begin{cases} \Pi_2 - \xi, & \text{if } I \text{ acquires } E, \\ \Pi_I + \Pi_E, & \text{otherwise,} \end{cases}$$

where $\xi \geq 0$ denotes the social cost of eliminating the entrant as an independent firm.

Proposition 8. *The acquisition is harmful whenever*

$$V_2 - 2V_1 \geq \xi.$$

This condition is equivalent to

$$k \leq \tilde{k}(n, \xi) \equiv \frac{n+1}{2} \frac{\sqrt{\xi(8n+\xi)} - 3\xi}{n - \xi},$$

which is U-shaped in n and increasing in ξ.

The social planner compares the economies of covariate integration reflected in $V_2 - 2V_1$ and the spillovers ξ to evaluate whether the loss of spillovers is worth the gains in prediction value it generates. However, the incumbent does not internalize the loss of positive spillovers, generating a tension between private and social incentives for acquisition. Although scope economies make data combinations privately valuable, they also cause the incumbent to systematically over-acquire entrants. The incumbent captures all private complementarities but does not internalize the entrant's contribution to future innovation, data generation, or technological diversity. The resulting acquisitions are therefore privately efficient but potentially socially harmful—too many mergers occur from society's perspective.

Regulation and Remedies Privacy regulation can worsen this tension. By raising compliance costs, the EU's General Data Protection Regulation (GDPR) lowers k, thereby dampening economies of scope and reducing the interval $[0, \tilde{k}(n, \xi)]$ in which the acquisition is socially desirable. Conversely, open-data initiatives raise k and broaden the interval in which economies of scope make acquisition desirable. These results suggest that privacy law, data governance, and merger policy must be coordinated: measures that restrict data access may unintentionally increase the likelihood of harmful data-driven acquisitions.

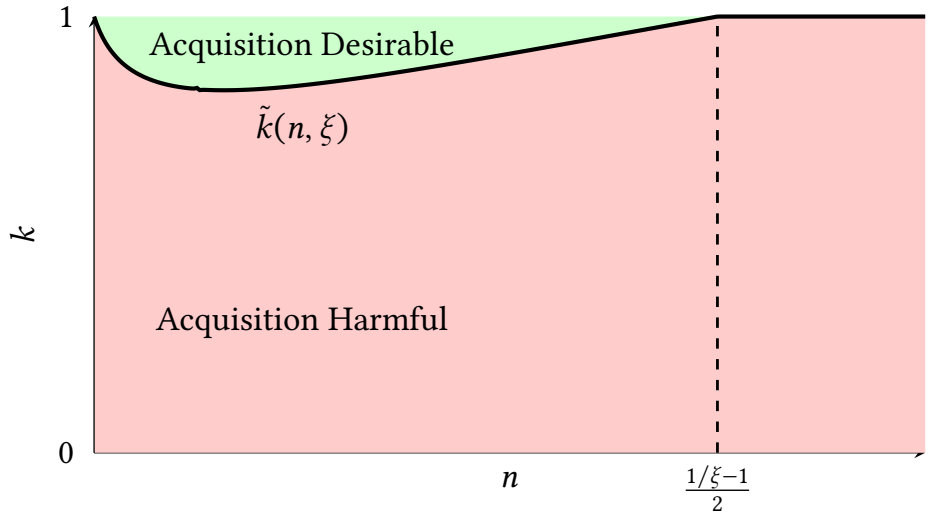

Figure 2: An acquisition is harmful for all (n, k) below the U-shaped threshold $k = \tilde{k}(n, \xi)$, where the loss of innovation spillovers exceeds the scope economies from combining datasets.

A potential remedy to potentially harmful acquisitions are FRAND APIs (Fair, Reasonable, and Non-Discriminatory Application Programming Interfaces). FRAND APISs are a regulated mechanism that guarantees standardized, non-exclusive access to a firm's data on fair technical and pricing terms. It preserves interoperability by allowing rivals to continue accessing the entrant's data after a merger. In this model, a FRAND API mitigates the dynamic loss ξ by keeping the entrant's data and experimentation available to the ecosystem, while still allowing the incumbent to capture the scope economies $\Delta(n, k)$. Therefore, the model provides support to the FRAND access obligations adopted in EU digital regulation as merger remedies, most notably in the Digital Markets Act and the Data Act.

5.2 Data Pools

Data owners often form partnerships to pool their datasets and sell access jointly. For example, BMW, Mercedes-Benz, and Audi co-founded the platform *Here Mobility Data Marketplace*, which aggregates GPS, speed, and road-condition data from connected cars. These agreements can benefit society as they address an instance fo the fundamental *complements problem* originally noted by **cournot1838**: if independent monopolists sell datasets that are complements, the resulting double marginalization leads to inefficiently high prices. Consequently, policymakers have encouraged pooling or sharing of datasets to overcome these frictions in the EU Data Act and in the European strategy for common data spaces. However, several papers have warned that brokers might use data sharing agreements to collude.
To assess the merits of these arguments, we develop a model in the spirit of Lerner and Tirole (2004): it turns out that the economics of combining datasets closely matches the economics of combining patents.

5.2.1 Setup

Data Owners Consider two data owners, each holding a dataset of identical informativeness. The complete dataset consists of n observations and k covariates, with the training and prediction covariates coinciding. However, the data may be split between the two owners either along the n (sample size) dimension or along the k (covariate) dimension. All parties are symmetrically informed about the informativeness of each dataset.
If the data are split along the sample dimension, each owner holds n observations and all k covariates. Pooling thus doubles the sample size. Conversely, if the data are split along the covariate dimension, each owner holds all n observations but only half the covariates, so pooling expands the covariate space. These two cases capture two distinct sources of complementarity: statistical precision (more n) and informational richness (more k).

Data Buyers There is a continuum of potential buyers (e.g., prediction firms) who can purchase access to one or both datasets and combine them without cost. Buyers are heterogeneous and indexed by $\theta \in [\underline{\theta}, \bar{\theta}]$, representing their adoption cost or opportunity cost of using the prediction technology.
A buyer of type θ who purchases access to $q \in \{1, 2\}$ datasets obtains a gross surplus

$$U_q = V_q - \theta,$$

where V_q denotes the predictive value of having access to q datasets. Specifically,

$$V_q \equiv \begin{cases} A\left(\dfrac{2n}{q},\ k\right), & \text{if the data are split along the } n \text{ dimension,} \\ A\left(n,\ \dfrac{2k}{q}\right), & \text{if the data are split along the } k \text{ dimension.} \end{cases}$$

Since $V(n, k)$ is increasing in both arguments, combining datasets strictly improves predictive value. Intuitively, pooling along the n dimension increases the number of observations available for training, which reduces estimation error, whereas pooling along the k dimension increases the number of predictive features, which broadens the scope of prediction.
The heterogeneity parameter θ is distributed according to

$$G(\theta) = \theta^{\alpha}, \qquad \alpha \in [0, 1],$$

implying a strictly increasing hazard rate $\frac{g(\theta)}{1-G(\theta)}$ with $g = G'$. The parameter α thus governs the curvature of demand: when α is low, heterogeneity is large and demand is relatively inelastic; when α is high, buyers are more homogeneous and demand becomes more elastic.
Given a price P for access to a pool of $q \in \{1, 2\}$ datasets, only buyers with $V_q - \theta \geq P$ make

the purchase. Hence, the corresponding demand function is

$$D(P; V_q) = \Pr(V_q - \theta \geq P) = (V_q - P)^{\alpha}.$$

This formulation implies that the elasticity of demand increases in α: higher α corresponds to a market in which adoption falls faster as price rises. Equivalently, α can be interpreted as the *semi-elasticity of demand*, describing how responsive adoption is to changes in price.

5.2.2 Pooling Price

When data owners form a pool, they coordinate pricing and behave as a single monopolist offering a pooled dataset. This situation mirrors the *pool pricing benchmark* in Section I.b of Lerner and Tirole (2004), where a patent pool chooses the package price P to maximize joint revenue given demand $D(P - V_2)$. The logic is analogous here: the pooled data pool yields predictive value V_2, and all buyers face a single posted price P.

Lemma 3 (Optimal Pool Price)**.** *The optimal pool price is*

$$P^* = \frac{V_2}{\alpha + 1}.$$

The pool acts as a monopolist setting the joint-access price that equates marginal revenue to zero, just as in Equation (1) in Lerner and Tirole (2004). The resulting price is decreasing in the demand semi-elasticity α: when α is high, buyers are more sensitive to price, leading the pool to charge less. Conversely, when α is low, demand is inelastic, so the pool can extract a larger fraction of the total value V_2.
Economically, pooling internalizes the complementarity between the dataset, analogous to Cournot's double marginalization problem in complementary goods. When data are sold separately, each owner fails to account for the positive externality that lower prices have on the other's sales. By setting a single joint price, the pool eliminates this inefficiency and behaves as an integrated monopolist.

5.2.3 Covariate Silos

If the datasets are split in two sets of covariates, the datasets are complements by Corollary **??**. We can use the results in Section II in Lerner and Tirole (2004) to characterize the unique symmetric equilibrium in the case in which brokers selling complementary data do not form a pool.

Lemma 4 (Covariate Fragmentation Price)**.** *If the brokers have distinct covariates each B_i prices at*

$$p_i = \frac{V_2}{2 + \alpha},$$

and the buyers will buy from both brokers.

Applying Proposition 1 in Lerner and Tirole (2004) directly yields the following result.

Proposition 9. *A pool of buyers with distinct covariates is always procompetitive.*

Covariate silos are an instance of the classic complements problem of **cournot1838**. Each broker prices without internalizing that lowering its price increases the value of the other dataset. Pooling eliminates this double marginalization and behaves exactly like integration in complementary patents in Lerner and Tirole (2004).

5.2.4 Observation Silos

If the datasets are split in two sets of covariates, the datasets are substitutes by Corollary ??. We can use the results in Section II in Lerner and Tirole (2004) characterizes the unique symmetric equilibrium in the case in which the brokers selling substitute data do not form a pool.

Lemma 5 (Sample Fragmentation Price)**.** *If the brokers have different observations on the same covariates,*

$$p_i = \min\left\{V_2 - V_1, \frac{V_2}{2+\alpha}\right\},$$

and the buyers will buy from both brokers.

Applying Proposition 1 in Lerner and Tirole (2004) directly yields the following characterization of when fragmented sample pooling is procompetitive.

Proposition 10 (Welfare of Observation Pooling)**.** *A pool of observations is procompetitive if and only if*

$$k < \hat{k}(n,\alpha) \equiv 1 - \frac{n}{2\alpha}.$$

The condition $k \leq \hat{k}(n,\alpha)$ mirrors the distinction between the *competition margin* and the *demand margin* in Lerner and Tirole (2004). When k is low, the incremental predictive value of the second block of observations, $V_2 - V_1$, remains substantial. Buyers therefore strictly prefer to acquire both datasets, so each broker must set a price low enough not to be excluded. Fragmented pricing is thus constrained by the competition margin, and the two datasets act as effective complements at the equilibrium prices. Separate pricing then reproduces the classic Cournot double-marginalization problem, and pooling internalizes this externality, lowering the total price.

When $k > \hat{k}(n,\alpha)$, the incremental value $V_2 - V_1$ becomes small. Buyers are nearly indifferent between acquiring one or two datasets, so each broker can raise its price without being displaced. Pricing is now governed by the demand margin, and a pool relaxes this competitive pressure. In this region, pooling raises the joint-access price and is therefore anti-competitive.

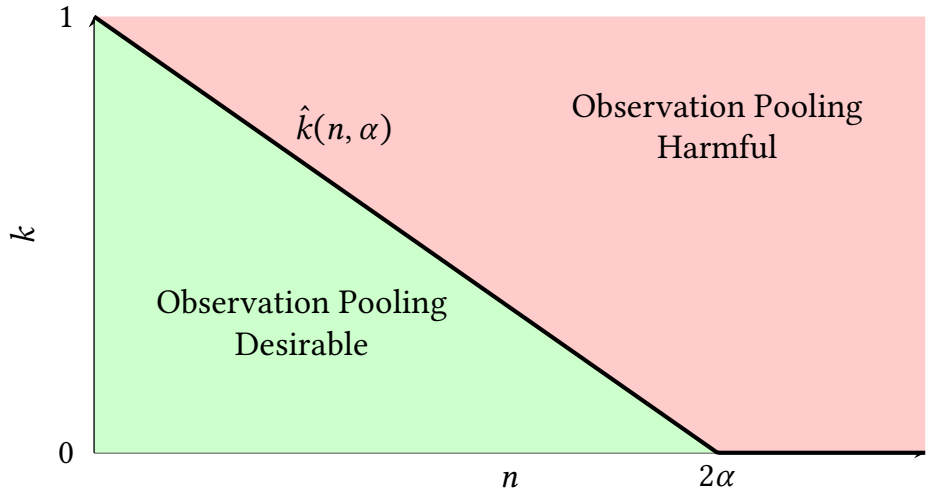

Figure 3: The threshold $k = \hat{k}(n, \alpha)$ separates the region in which fragmented pricing is constrained by the competition margin (pooling lowers prices) from the region in which it is constrained by the demand margin (pooling raises prices).

Because $\hat{k}(n, \alpha)$ is decreasing in n, the model offers a sharp statistical insight: observation pooling becomes anti-competitive when datasets are large. As n grows, posterior variance is already small, so doubling the sample yields only a negligible reduction in prediction error. The marginal value $V_2 - V_1$ of the second dataset therefore collapses. In this case, fragmentation disciplines prices through the competition margin—each broker must keep its price close to $V_2 - V_1$ to avoid being dropped—while pooling removes this discipline. As a result, pooling observations raises the price of access and is anti-competitive in the upper-right corner of the (n, k) plane.

Regulation of Data Pools The model qualifies current policy discussions on data sharing. European legislation such as the EU Data Act and the European strategy for common data spaces broadly promotes data pooling and data spaces, which can overcome the Cournot complements problem in fragmented data markets. However, these policy initiatives do not distinguish between sources of dataset heterogeneity (covariates vs. observations) as a condition for pooling to be procompetitive. The model shows that this distinction is essential. Pooling heterogeneous user attributes (different covariates) always resolves a pure complements problem and is therefore procompetitive, fully consistent with the intuition in the Data Act Recitals that data portability and interoperability reduce market power. In contrast, pooling additional observations on the same attributes may be anticompetitive when $k > \hat{k}(n, \alpha)$: in this region the second dataset adds little predictive value, fragmented pricing would be disciplined by the competition margin, and a pool merely relaxes competitive pressure.
These results imply that policies encouraging sharing of data on similar attributes may unintentionally raise prices in markets where data is abundant. Conversely, when different at-

tributes are shared, the model confirms the procompetitive rationale underlying EU data-space initiatives. More generally, the model highlights the need for analytical tools within merger control and data-access regulation to understand the level of heterogeneity of the shared data, a distinction absent from existing regulatory guidance.

5.3 Data Exclusivity

Recent exclusive data-licensing agreements—such as Reddit's 2024 deals with OpenAI and Google—highlight broader concerns that proprietary access to datasets may distort competition in AI and prediction markets. Because data are non-rival, a data seller faces a dynamic commitment problem akin to that of a durable-good monopolist: once it has licensed the dataset to one firm, it is tempted to also license it to the rival, eroding the first buyer's advantage. As in Katz and Shapiro (1986) and Aghion and Bolton (1987), exclusivity can serve as a contractual commitment device that mitigates opportunism by the seller but introduces a welfare trade-off: it softens business-stealing while depressing investment by excluded firms when data and proprietary inputs are complements. The model predicts that profitable exclusivity may become socially undesirable when datasets are abundant and product-market rivalry is intense. In such settings, exclusivity amplifies incumbency advantages and can deter entry—offering a micro-foundation for regulatory scrutiny of data-sharing agreements such as the Reddit–OpenAI deal.

5.3.1 Setup

There are three players: a data seller S (e.g., a platform holding user-generated data) and two prediction firms F_1 and F_2. The firms compete to sell predictions to customers whose utility depends on the value of the prediction.

Prediction Buyers. There is a unit mass of customers, divided into:

- a mass $s \in [0, 1]$ of *shoppers*, who can buy from either firm;
- a mass $(1 - s)/2$ of *captive customers* for each firm, who can only buy from that firm.

The parameter s captures the intensity of competition: when $s = 0$, all customers are captive and firms behave as local monopolists; when $s = 1$, all customers are shoppers and the market is fully competitive. This simple structure captures the idea that data quality matters only in relative terms, since shoppers migrate toward the firm offering the more accurate prediction. Customer utility from firm $i \in \{1, 2\}$ is

$$u_i = V_i - p_i,$$

where V_i denotes the acccuracy of the prediction and p_i is the price charged. Shoppers buy from the firm with the highest net utility $V_i - p_i$, while captive customers always purchase from their incumbent firm.

Data Buyers. We assume covariates are ex-ante identical and that training and prediction covariates coincide. Each firm $i \in \{1, 2\}$ starts with no data and can improve its prediction quality through two channels:

1. *Training Data Licensing.* Firm i can buy a license for the seller's training dataset of k covariates and the realization of the target variable for n observations, paying a fee f. This choice is represented by a binary variable $\ell_i \in \{0, 1\}$.

2. *Proprietary Data Collection.* Firm i can collect the k covariates for the target user, paying a fixed cost $c > 0$, represented by a binary choice $r_i \in \{0, 1\}$.

A firm's data strategy is $(\ell_i, r_i) \in \{0, 1\}^2$, the prediction quality of firm i is

$$V_i \equiv V(\ell_i, r_i) = \hat{V} \times \mathbf{1}\left(\ell_i r_i = 1\right),$$

where

$$\hat{V} \equiv \frac{k}{\frac{1-k}{n} + 1}.$$

Intuitively, in order to make a prediction, the firm must license the training data and invest to collect proprietary data on the target individual.

Pricing and Profits. Firms can price discriminate between captive customers and shoppers. Denote prices by p_i^c (captives) and p_i^s (shoppers). Let D_i^s denote firm i's share of shoppers, with $D_1^s + D_2^s = 1$. Firm i's revenue from prediction is then

$$\pi_i = p_i^c \frac{1-s}{2} + p_i^s s D_i^s,$$

and total profit is

$$\Pi_i = \pi_i - c\, r_i - f\, \ell_i.$$

The cost terms capture the trade-off between acquiring data and improving prediction quality: collecting proprietary data entails the fixed cost c, while licensing requires the payment f to the data seller.

Data Seller. The data seller S has a monopoly over the training dataset and sets a take-it-or-leave-it license fee f.[9] The seller's profit is

$$\Pi_S = f \sum_{i \in \{1,2\}} \ell_i.$$

Because data are non-rival, S could in principle sell to both firms without loss of quality, but doing so may erode the exclusivity premium paid by the first buyer. This creates the central commitment problem: after selling to one firm, the seller is tempted to license to the rival as well, thereby reducing the first buyer's willingness to pay ex ante.

Social Planner. A benevolent social planner evaluates total welfare as

$$W = \Pi_S + \Pi_1 + \Pi_2 + CS,$$

where customer surplus is given by

$$CS = \sum_{i \in \{1,2\}} \left[(V_i - p_i^c) \frac{1-s}{2} + (V_i - p_i^s) s D_i^s \right].$$

This allows us to assess how exclusivity affects welfare through both prices and investment incentives.

Timing. Information is complete. The game unfolds in four stages:

1. **Data Pricing.** The data seller S sets a publicly observed license fee f;
2. **Data Collection.** Each firm F_i chooses $(\ell_i, r_i) \in \{0,1\}^2$, which are publicly observed;
3. **Prediction Pricing.** Firms simultaneously set (p_i^c, p_i^s). Customers choose the firm offering higher utility,
4. **Realization.** Nature draws the data and the target variable, prediction losses and profits are realized for S, F_1, and F_2.

We solve the game by backward induction and characterize the Subgame Perfect Equilibrium (SPE).

5.3.2 Prediction Pricing

We analyze the pricing subgame, taking prediction accuracies (V_1, V_2) as given from the data collection. Firms can price-discriminate between captive customers and shoppers.

[9] Allowing S to set different fees for F_1 and F_2 would not affect the analysis as far s exclusivity is concerned.

Captive pricing. Let the outside option yield utility 0. Assuming they are expected utility maximizers, captive customers of F_i buy if and only if $V_i - p_i \geq 0$. Therefore, the unique optimal price is

$$p_i^c = V_i \qquad \text{for each } i \in \{1, 2\}. \tag{7}$$

The firm extracts the full surplus of captives.

Shopper pricing. Shoppers buy from the firm that offers the higher net utility; if $V_i - p_i > V_j - p_j^s$ they all buy from F_i, if $V_i - p_i < V_j - p_j^s$ they all buy from F_j, and if equal any split yields the same payoff for firms and customers.[10] Write $(x)^+ \equiv \max\{x, 0\}$. The next lemma characterizes the unique trembling-hand perfect equilibrium for shoppers.

Lemma 6 (Shopper-price equilibrium)**.** *Fix (V_1, V_2). The pricing subgame in shopp[Shopper-price equilibrium]er prices admits a unique equilibrium under trembling-hand perfection:*

$$p_i^s = (V_i - V_j)^+, \qquad i \neq j, \tag{8}$$

so that (i) if $V_i > V_j$, firm i sets $p_i^s = V_i - V_j$, firm j sets $p_j^s = 0$, and all shoppers buy from i; (ii) if $V_i = V_j$, both set $p_i^s = p_j^s = 0$ and shoppers can be split arbitrarily.

Combining Equation (7) and Lemma 6, the equilibrium profit of firm i given data strategies (ℓ_i, r_i) and (ℓ_j, r_j) (which pin down (V_i, V_j)) is

$$\pi(\ell_i, r_i; \ell_j, r_j) \;=\; \frac{1-s}{2} V_i \;+\; s\,(V_i - V_j)^+, \qquad j \neq i. \tag{9}$$

The first term is captive revenue at the value price; the second term is shopper revenue, which equals the quality advantage when the firm is better and zero otherwise.

5.3.3 Cost Regions

The trade-off between exclusivity and non-exclusivity depends on the equilibirum of the proprietary data investment subgame, which in turn depends on the cost of collecting proprietary data c relative to the value $\hat{V}$. Two regimes arise:

- **Low-cost region:** $c \in [0, \bar{c})$, where

$$\bar{c} \equiv \frac{1-s}{2}\hat{V}.$$

  Here proprietary data are cheap enough that any licensed firm will always invest.

[10] This tie-breaking rule is without loss for the equilibrium characterization; a trembling-hand refinement will select the limit outcome stated below.

- **High-cost region:** $c \in [\bar{c}, \bar{\bar{c}})$, where

$$\bar{\bar{c}} \equiv \frac{1+s}{2}\hat{V}.$$

  Here investment occurs only when a firm obtains a monopoly over shoppers, leading to mixed strategies when both license.

- **No investment region:** $c \geq \bar{\bar{c}}$. Here investment never occurs.

These regions generate two distinct exclusivity incentives for the seller and two distinct welfare benchmarks for the planner, which we will analyze separately.

5.3.4 Low-cost Region

Proprietary Data Investment. If $c < \bar{c}$, the cost of investing is less than the increase in revenue on a firm's captive customers. Therefore investment will occur if and only if the license is purchased.

Proposition 11. *If $c < \bar{c}$, for any licensing choice (ℓ_1, ℓ_2) there is a unique Nash Equilibrium*

$$(r_1^*, r_2^*) \;=\; (\ell_1, \ell_2).$$

The intuition is straightforward: the rival's competition in the shopper segment is irrelevant because the incentives to invest in the captive segment are sufficiently large relative to the cost of investing. Furthermore, because investing in proprietary data is costly but gives no benefit if the license is acquired, in case a firm does not license it will also avoid investing.

Data Pricing. The seller chooses these fees to maximize its profit, taking into account that the fees determine how many firms choose to license the data. The following result characterizes the maximal fee f_d compatible with $d \in \{1, 2\}$ firms purchasing the data.

Lemma 7. *If $c < \bar{c}$,*

$$f_2 = \frac{1-s}{2}\hat{V} - c, \quad \textit{and} \quad f_1 = \frac{1+s}{2}\hat{V} - c.$$

Intuitively, if both firm license the training data they will both invest and will only be able to extract the surplus of the captive customers $\frac{1-s}{2}\hat{V}$. Instead, if only one licenses, it will extract the full surplus of the shoppers as well, giving it an additional demand of sA, because by Proposition 11 it will invest, and since $f_1 > f_2$ the rival will not license as it is only profitable for a single firm to license.

If the profit from selling to one firm is higher than selling to both, i.e., $f_1 > 2f_2$, the data seller will pursue *de facto exclusivity*, setting a price so high hat only one firm will choose to purchase the data and the other will avoid entry.

Proposition 12. *The data seller will choose $f^* = f_1$, i.e., it will pursue de facto exclusivity, if $c \in [\underline{c}, \bar{c})$, where*

$$\underline{c} \equiv \frac{1-3s}{2}\hat{V} \in [0, \bar{c}].$$

Therefore, whenever competition is fierce, i.e. $s > 1/3$, the seller will always prefer de facto exclusivity. If $s \leq 1/3$, the seller will prefer de facto exclusivity if the cost of collecting prediction covariates is large compared to $\hat{V}$. Since $\hat{V} = V(n, k)$, which is increasing in both arguments, exclusivity is profitable if datasets are sufficiently small.

Social Welfare. I assume the planner can choose between de facto exclusivity and real non exclusivity but cannot influence the proprietary data investment equilibrium.

Proposition 13. *If $c < \bar{c}$, the planner prefers real non-exclusivity to de facto exclusivity.*

The cost is too low for the motive of avoiding excessive entry and overinvestment in data collection ot make it socially desirable to pursue de facto exclusivity. The following result characterizes an interval in which de facto excusivity is profitable for the data seller even if it harms social welfare.

Corollary 9. *The seller chooses de facto exclusivity even though it is harmful if and only if $c \in (\underline{c}, \bar{c})$.*

The length of the interval is $\bar{c} - \underline{c} = s\hat{V}$, so harmful exclusivity is more likely if competition is fierce and, given $\hat{V} \equiv V(n, k)$, which is increasing in both arguments, if data is abundant.

5.3.5 High-cost Region

I now consider what happens when the cost of proprietary data collection is $c \in (\bar{c}, \bar{\bar{c}})$. This means the cost of investing is more than the surplus of a firm's captive customers but possibly less than the cumulative surplus of a firm's captive customers and the shoppers.

Proprietary Data Investment. If $c \in (\bar{c}, \bar{\bar{c}})$, a firm will collect prediction covariates with probability one only if it can sell as a monopolist to the shoppers, meaning that it benefits from de facto exclusivity. If both firms purchase the license, there is no pure-strategy equilibrium in the proprietary data-collection subgame.

Proposition 14. *We distinguish two subcases:*

- ***De facto exclusivity:*** *Without loss of generality, let $\ell_1 = 1$, $\ell_2 = 0$, the unique NE is*

$$r_1 = 1, \qquad r_2 = 0.$$

- ***Both firms license:*** $\ell_1 = \ell_2 = 1$. *The unique symmetric mixed strategy equilibrium has each firm investing with probability*

$$\xi^* = \frac{V(1+s) - 2c}{2As}, \qquad \xi^* \in [0, 1].$$

In equilibrium, profits are zero: $\Pi_i^{\text{mix}} = 0$.

The mixed strategy equilibrium investment probability is increasing in s if and only if $c > \hat{V}/2$ (i.e., when proprietary data are relatively costly) and decreasing in s when $c < \hat{V}/2$.

Data Pricing. As the profit of the mixed strategy equilibirum is zero, firms will have no willingness to pay for the license unless there is de facto exclusivity. Under de facto exclusivity, only one firm licenses and invests, while the rival neither licenses nor invests.

Proposition 1. *If $c \in (\bar{c}, \bar{\bar{c}})$, the data seller will always choose $f^* = f_1$, i.e., it will always pursue de facto exclusivity, where*

$$f_1 = \frac{1+s}{2} V_2 - c.$$

Social welfare. The following result characterizes the social planner optimum, by comparing the pure strategy equilibrium of the investment subgame and the mixed strategy non-exclusive one.

Proposition 15. *If $c \in (\bar{c}, \bar{\bar{c}})$, the planner prefers de facto exclusivity to real non-exclusivity.*

The overall investment in the real non exclusive case is so low that the planner prefers to gran exclusivity in order to ensure that investment occurs.

5.3.6 Summary.

The equilibrium interaction between exclusivity, investment, and welfare is governed by the fixed cost of proprietary data collection c, relative to the value

$$\hat{V} \equiv \frac{k}{\frac{1-k}{n} + 1}.$$

Three cost thresholds determine firms' willingness to collect proprietary covariates:

$$\underline{c} \equiv \frac{1-3s}{2}\hat{V}, \qquad \bar{c} \equiv \frac{1-s}{2}\hat{V}, \qquad \bar{\bar{c}} \equiv \frac{1+s}{2}\hat{V}.$$

For fixed (c, s) we can invert these equations and obtain three endogenous thresholds in covariate richness,

$$\underline{k}(n, c, s), \qquad \bar{k}(n, c, s), \qquad \bar{\bar{k}}(n, c, s),$$

defined implicitly by

$$c = \frac{1-3s}{2} V(n, \underline{k}), \qquad c = \frac{1-s}{2} V(n, \bar{k}), \qquad c = \frac{1+s}{2} V(n, \bar{\bar{k}}).$$

For $s < 1/3$ these satisfy

$$\underline{k}(n, c, s) < \bar{k}(n, c, s) < \bar{\bar{k}}(n, c, s) \qquad \text{for all } (n, c).$$

These thresholds partition the (n, k)-space into four regions:

- **No investment:** If $k < \underline{k}(n, c, s)$, proprietary data collection is never profitable, even under exclusivity. No firm invests, and licensing decisions have no effect.
- **Beneficial exclusivity:** If $k \in [\underline{k}(n, c, s), \bar{k}(n, c, s))$, investment occurs only under exclusivity, because a firm collects proprietary data if and only if it can monopolize shoppers. Both the seller and the planner prefer exclusivity, as it ensures that at least one firm invests.
- **Harmful exclusivity:** If $k \in [\bar{k}(n, c, s), \bar{\bar{k}}(n, c, s))$, both firms would invest even without exclusivity, but the seller prefers to restrict access to extract a higher fee. The planner prefers non-exclusivity to keep both firms active. Exclusivity is privately profitable but socially harmful.
- **No exclusivity:** If $k \geq \bar{\bar{k}}(n, c, s)$, proprietary data are sufficiently cheap relative to dataset quality that both firms always invest. Both the seller and the planner prefer non-exclusivity, so broad data access is jointly optimal.

The figure below plots these three thresholds and the resulting four regions for a fixed pair (c, s) with $s < 1/3$.

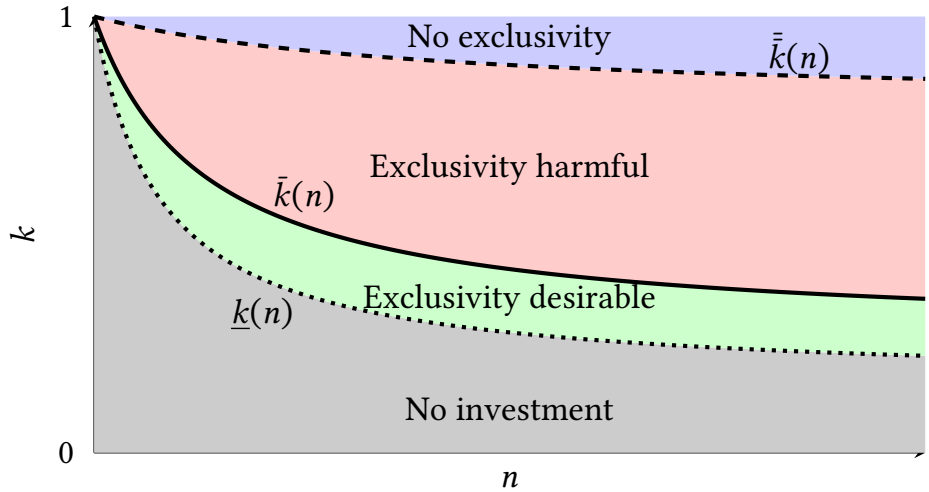

Figure 4: The three thresholds $k = \underline{k}(n)$, $k = \bar{k}(n)$, and $k = \bar{\bar{k}}(n)$ partition the (n, k)-space into four regions. Below $\underline{k}(n)$, investment never occurs. Between $\underline{k}(n)$ and $\bar{k}(n)$, exclusivity is jointly optimal because it induces investment. Between $\bar{k}(n)$ and $\bar{\bar{k}}(n)$, exclusivity is privately profitable but socially harmful. Above $\bar{\bar{k}}(n)$, both the seller and the planner prefer non-exclusivity.

6 Conclusion

This paper develops a general framework for understanding the value of data in prediction by explicitly modeling covariates. The analysis shows how economies of scope across covariates, interactions between covariates and observations, and complementarities between training and prediction can generate increasing returns, offering a microfoundation for the rich-get-richer effects often observed in data-driven markets.

These forces have direct implications for policy and strategy. Prediction technologies may display natural monopsony characteristics, as concentrating covariates within one firm can raise efficiency. Privacy regulation that fragments data supply may inadvertently reinforce monopsony power, creating a trilemma between privacy, competition, and efficiency. The framework also highlights that not all data pooling agreements are alike: pooling lists of users with the same covariates can be anticompetitive, whereas pooling different covariates on similar users raise welfare by eliminating double marginalization. Exclusivity deals, such as those signed between AI labs and data providers, may profitably foreclose entry by depriving rivals of essential complements. For firms, the results imply that prediction entails substantial sunk costs: early on, investment should balance user acquisition and attribute enrichment, while specialization and integration become optimal at a larger scale.

More broadly, the analysis cautions against treating data as homogeneous. Policies promoting open data without regard to dataset composition may miss crucial efficiency margins, whereas

access remedies such as FRAND-priced APIs or federated learning preserve economies of scope.

My work opens two natural avenues for future research. The first is empirical. I aim to develop a methodology to test my results on real datasets. While the existing empirical literature[11] provides partial support to my findings, it suffers from two limitations: (i) most studies focus on a single dataset, whereas uncovering general properties requires comparing multiple datasets along common dimensions; and (ii) no existing work systematically tests all the properties identified in my model. Once these empirical properties are validated, my framework could serve as the foundation for a practical formula for data valuation, in the spirit of the Black–Scholes–Merton formula for derivatives.[12] The second avenue is theoretical. Embedding my static model into a dynamic Wald sampling framework would allow me to microfound data-enabled learning and analyze when feedback loops generate convergent data-collection strategies versus when they diverge.

[11] See Bajari et al. (2019), Schaefer and Sapi (2023), Lee and Wright (2023), Yoganarasimhan (2020), and Carballa-Smichowski et al. (2025b)

[12] See Black and Scholes (1973), Merton (1973)

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Proofs

Proposition 1 (Optimal predictor)**.** *The optimal predictor is*

$$\hat{y}^*(\boldsymbol{D}) = \mathbb{E}\left[y \mid \boldsymbol{D}\right] = \sum_{k \in \mathcal{P} \cap \mathcal{T}} x_{0,k} \mathbb{E}\left[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right].$$

Proof. Fix a design $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$ and a realized dataset $\boldsymbol{D} = ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}})$. Under quadratic loss, the posterior risk of a predictor $\hat{y}(\boldsymbol{D})$ is

$$\rho_{\boldsymbol{D}}(\hat{y}) = \mathbb{E}\left[(y_0 - \hat{y}(\boldsymbol{D}))^2 \mid \boldsymbol{D}\right].$$

For any $\boldsymbol{D}$, expand the square around the posterior mean:

$$\begin{aligned}\mathbb{E}\left[(y_0 - \hat{y})^2 \mid \boldsymbol{D}\right] &= \mathbb{E}\left[(y_0 - \mathbb{E}[y_0 \mid \boldsymbol{D}] + \mathbb{E}[y_0 \mid \boldsymbol{D}] - \hat{y})^2 \mid \boldsymbol{D}\right] \\ &= \mathbb{E}\left[(y_0 - \mathbb{E}[y_0 \mid \boldsymbol{D}])^2 \mid \boldsymbol{D}\right] + (\hat{y} - \mathbb{E}[y_0 \mid \boldsymbol{D}])^2,\end{aligned}$$

where the cross term is zero because $\mathbb{E}[y_0 - \mathbb{E}[y_0 \mid \boldsymbol{D}] \mid \boldsymbol{D}] = 0$. Hence $\rho_{\boldsymbol{D}}(\hat{y})$ is uniquely minimized by

$$\hat{y}^*(\boldsymbol{D}) = \mathbb{E}[y_0 \mid \boldsymbol{D}].$$

It remains to compute $\mathbb{E}[y_0 \mid \boldsymbol{D}]$. By the DGP,

$$y_0 = \sum_{k=1}^{Z} \beta_k x_{0k} + \eta_0,$$

with η_0 independent of $\boldsymbol{D}$ and mean zero, so $\mathbb{E}[\eta_0 \mid \boldsymbol{D}] = 0$ and

$$\mathbb{E}[y_0 \mid \boldsymbol{D}] = \sum_{k=1}^{Z} x_{0k}\, \mathbb{E}[\beta_k \mid \boldsymbol{D}].$$

Moreover, conditional on $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$, the posterior of $\boldsymbol{\beta}$ does not depend on $\boldsymbol{x}_{0,\mathcal{P}}$, since $\boldsymbol{x}_{0,\mathcal{P}}$ is independent of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta}, \eta_0)$ under the prior. Thus $\mathbb{E}[\beta_k \mid \boldsymbol{D}] = \mathbb{E}[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})]$.
Finally, for $k \notin \mathcal{T}$, the training data are independent of β_k (because β_k only affects outcomes through x_{ik}, which is unobserved when $k \notin \mathcal{T}$), so the posterior equals the prior and $\mathbb{E}[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})] = \mathbb{E}[\beta_k] = 0$. Also, x_{0k} is observed only for $k \in \mathcal{P}$, so the sum reduces to $k \in \mathcal{P} \cap \mathcal{T}$:

$$\hat{y}^*(\boldsymbol{D}) = \mathbb{E}[y_0 \mid \boldsymbol{D}] = \sum_{k \in \mathcal{P} \cap \mathcal{T}} x_{0k}\, \mathbb{E}[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})].$$

□

Proposition 2 (Value of a dataset)**.** *The value of a dataset $\boldsymbol{D}$ is*

$$v(\boldsymbol{D}) = \operatorname{Var}[y] - \operatorname{Var}[y \mid \boldsymbol{D}] = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^{\top} \left(\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}] - \operatorname{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})]\right) \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

Proof. Fix a design $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$ and a realized dataset $\boldsymbol{D} = ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}})$. Under quadratic loss, Proposition 1 gives $\hat{y}^*(\boldsymbol{D}) = \mathbb{E}[y_0 \mid \boldsymbol{D}]$ and

$$\rho_{\boldsymbol{D}}(\hat{y}^*) = \min_{\hat{y}} \mathbb{E}[(y_0 - \hat{y})^2 \mid \boldsymbol{D}] = \mathrm{Var}(y_0 \mid \boldsymbol{D}).$$

Moreover, with no data the optimal constant predictor is $\hat{y}^*(\varnothing) = \mathbb{E}[y_0]$ and

$$\rho_{\varnothing}(\hat{y}^*(\varnothing)) = \min_{\hat{y}} \mathbb{E}[(y_0 - \hat{y})^2] = \mathrm{Var}(y_0).$$

Therefore,

$$v(\boldsymbol{D}) = \rho_{\varnothing}(\hat{y}^*(\varnothing)) - \rho_{\boldsymbol{D}}(\hat{y}^*) = \mathrm{Var}(y_0) - \mathrm{Var}(y_0 \mid \boldsymbol{D}).$$

It remains to express $\mathrm{Var}(y_0 \mid \boldsymbol{D})$. By the DGP,

$$y_0 = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} + \sum_{k \notin \mathcal{P}\cap\mathcal{T}} \beta_k x_{0k} + \eta_0,$$

and under the maintained assumptions: (i) η_0 is independent of $\boldsymbol{D}$ with $\mathrm{Var}(\eta_0) = \sigma_\eta^2 \tau^2$; (ii) $\boldsymbol{x}_{0,\mathcal{P}}$ is observed and thus fixed conditional on $\boldsymbol{D}$; (iii) for $k \notin \mathcal{T}$ the training likelihood does not depend on β_k, hence $\boldsymbol{\beta}_{\mathcal{P}\setminus\mathcal{T}}$ remains independent of the training data with posterior equal to prior and mean zero, so these coordinates do not affect $\mathbb{E}[y_0 \mid \boldsymbol{D}]$ (as shown in Proposition 1). For the conditional variance, note that conditioning on $\boldsymbol{D}$ fixes $\boldsymbol{x}_{0,\mathcal{P}}$, and η_0 is independent of $\boldsymbol{D}$, so

$$\mathrm{Var}(y_0 \mid \boldsymbol{D}) = \mathrm{Var}\big(\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid \boldsymbol{D}\big) + \mathrm{Var}(\eta_0).$$

Since $\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}$ is $\boldsymbol{D}$-measurable,

$$\mathrm{Var}\big(\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid \boldsymbol{D}\big) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}))\, \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

Unconditionally,

$$\mathrm{Var}(y_0) = \mathrm{Var}\big(\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}\big) + \mathrm{Var}(\eta_0) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}})\, \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}} + \mathrm{Var}(\eta_0),$$

where we used $\mathbb{E}[\boldsymbol{\beta}] = \boldsymbol{0}$ and independence across coordinates.
Subtracting the two expressions cancels $\mathrm{Var}(\eta_0)$ and yields

$$v(\boldsymbol{D}) = \mathrm{Var}(y_0) - \mathrm{Var}(y_0 \mid \boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \big(\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}) - \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}))\big) \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}},$$

which is the claimed formula. □

Corollary 1 (No useless prediction covariates)**.** *In any optimal design it must hold that* $\mathcal{P} \subseteq \mathcal{T}$.

Proof. Suppose by contradiction that an optimal design $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$ satisfies $\mathcal{P} \nsubseteq \mathcal{T}$. Let

$$\mathcal{P}' \equiv \mathcal{P} \cap \mathcal{T}, \qquad \mathcal{D}' \equiv (N, \mathcal{T}, \mathcal{P}').$$

By Proposition 4, for any realized dataset under design $\mathcal{D}$, the value depends on $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ and $\boldsymbol{x}_{0,\mathcal{P}}$ only through $\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}} = \boldsymbol{x}_{0,\mathcal{P}'}$. Hence the random dataset values coincide under $\mathcal{D}$ and $\mathcal{D}'$, and therefore

$$V(\mathcal{D}) = V(\mathcal{D}').$$

On the other hand, since $\mathcal{P}' \subsetneq \mathcal{P}$ and $C_{\mathcal{P}}(\cdot)$ is strictly increasing and positive,

$$C(\mathcal{D}') = C_N\left(\frac{N}{Z}\right) + C_{\mathcal{T}}\left(\frac{|\mathcal{T}|}{Z}\right) + C_{\mathcal{P}}\left(\frac{|\mathcal{P}'|}{Z}\right) < C_N\left(\frac{N}{Z}\right) + C_{\mathcal{T}}\left(\frac{|\mathcal{T}|}{Z}\right) + C_{\mathcal{P}}\left(\frac{|\mathcal{P}|}{Z}\right) = C(\mathcal{D}).$$

Thus,

$$V(\mathcal{D}') - C(\mathcal{D}') > V(\mathcal{D}) - C(\mathcal{D}),$$

contradicting the optimality of $\mathcal{D}$. Therefore, any optimal design must satisfy $\mathcal{P} \subseteq \mathcal{T}$. □

Lemma 1. *For any sequence of dataset designs $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ satisfying Assumption 1,*

$$(\varepsilon_1^{(Z)}, \dots, \varepsilon_N^{(Z)})^\top \underset{Z\to\infty}{\overset{d}{\to}} \mathcal{N}(\boldsymbol{0}_N, \sigma_\varepsilon^2(t)\ \tau^2 \cdot \boldsymbol{I}_N),$$

where the (relative) residual noise $\sigma_\varepsilon^2(t)$ *satisfies*

$$\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2.$$

Proof. Fix Z and write $T(Z) \equiv |\mathcal{T}(Z)|$. For each $i \in \{1, \dots, N(Z)\}$ define

$$\varepsilon_i^{(Z)} = \sum_{k=T(Z)+1}^{Z} \beta_k^{(Z)} x_{ik}^{(Z)} + \eta_i^{(Z)}.$$

Fix any integer $m \geq 1$ and consider the first m components $\boldsymbol{\varepsilon}_{[m]}^{(Z)} = (\varepsilon_1^{(Z)}, \dots, \varepsilon_m^{(Z)})^\top$.
Step 1 (conditional Gaussianity). Conditional on the coefficients $\boldsymbol{\beta}^{(Z)} = (\beta_1^{(Z)}, \dots, \beta_Z^{(Z)})$, the vectors $\boldsymbol{x}_k^{(Z)} \equiv (x_{1k}^{(Z)}, \dots, x_{mk}^{(Z)})^\top$ are i.i.d. across k with

$$\boldsymbol{x}_k^{(Z)} \mid \boldsymbol{\beta}^{(Z)} \sim \mathcal{N}\left(\boldsymbol{0}_m, \frac{\sigma_x^2}{Z}\boldsymbol{I}_m\right),$$

and are independent of $\boldsymbol{\eta}_{[m]}^{(Z)} = (\eta_1^{(Z)}, \dots, \eta_m^{(Z)})^\top$. Hence, conditional on $\boldsymbol{\beta}^{(Z)}$,

$$\sum_{k=T(Z)+1}^{Z} \beta_k^{(Z)} \boldsymbol{x}_k^{(Z)} \sim \mathcal{N}\left(\boldsymbol{0}_m, \frac{\sigma_x^2}{Z} \sum_{k=T(Z)+1}^{Z} (\beta_k^{(Z)})^2 \cdot \boldsymbol{I}_m\right),$$

and since $\boldsymbol{\eta}_{[m]}^{(Z)} \sim \mathcal{N}(\mathbf{0}_m, \sigma_\eta^2 \tau^2 \boldsymbol{I}_m)$ is independent, we obtain the conditional law

$$\boldsymbol{\varepsilon}_{[m]}^{(Z)} \mid \boldsymbol{\beta}^{(Z)} \sim \mathcal{N}\left(\mathbf{0}_m,\ s_Z^2\, \boldsymbol{I}_m\right), \qquad s_Z^2 \equiv \tau^2 \sigma_\eta^2 + \frac{\sigma_x^2}{Z} \sum_{k=T(Z)+1}^{Z} (\beta_k^{(Z)})^2.$$

Step 2 (limit of the random variance). By Assumption 1, $T(Z)/Z \to t$, so $(Z - T(Z))/Z \to 1-t$. By the law of large numbers,

$$\frac{1}{Z - T(Z)} \sum_{k=T(Z)+1}^{Z} (\beta_k^{(Z)})^2 \xrightarrow{p} \mathbb{E}[(\beta_1^{(Z)})^2] = \tau^2.$$

Therefore,

$$\frac{1}{Z} \sum_{k=T(Z)+1}^{Z} (\beta_k^{(Z)})^2 = \frac{Z - T(Z)}{Z} \cdot \frac{1}{Z - T(Z)} \sum_{k=T(Z)+1}^{Z} (\beta_k^{(Z)})^2 \xrightarrow{p} (1-t)\tau^2,$$

and thus

$$s_Z^2 \xrightarrow{p} \tau^2((1-t)\sigma_x^2 + \sigma_\eta^2) \equiv \tau^2 \sigma_\varepsilon^2(t).$$

Step 3 (convergence in distribution). For any fixed $\boldsymbol{w} \in \mathbb{R}^m$, the conditional characteristic function equals

$$\mathbb{E}\left[\exp\left(i \boldsymbol{w}^\top \boldsymbol{\varepsilon}_{[m]}^{(Z)}\right) \mid \boldsymbol{\beta}^{(Z)}\right] = \exp\left(-\tfrac{1}{2} s_Z^2 \|\boldsymbol{w}\|^2\right).$$

Taking expectations and using $s_Z^2 \to \tau^2 \sigma_\varepsilon^2(t)$ in probability,

$$\mathbb{E}\left[\exp\left(i \boldsymbol{w}^\top \boldsymbol{\varepsilon}_{[m]}^{(Z)}\right)\right] = \mathbb{E}\left[\exp\left(-\tfrac{1}{2} s_Z^2 \|\boldsymbol{w}\|^2\right)\right] \to \exp\left(-\tfrac{1}{2} \tau^2 \sigma_\varepsilon^2(t) \|\boldsymbol{w}\|^2\right),$$

which is the characteristic function of $\mathcal{N}(\mathbf{0}_m, \tau^2 \sigma_\varepsilon^2(t) \boldsymbol{I}_m)$. Hence,

$$(\varepsilon_1^{(Z)}, \dots, \varepsilon_m^{(Z)})^\top \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_m, \tau^2 \sigma_\varepsilon^2(t) \boldsymbol{I}_m\right).$$

Since m is arbitrary, this establishes that the residuals are asymptotically Gaussian with asymptotically independent components and variance $\tau^2 \sigma_\varepsilon^2(t)$, where $\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2$. □

Proposition 3 (Optimal Predictor). *The optimal predictor is*

$$\hat{y}^*(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \underbrace{\left(\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t) \cdot \boldsymbol{I}_{|\mathcal{T}|}\right)^{-1} \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{y}_N\right)_{\mathcal{P}\cap\mathcal{T}}}_{=\mathbb{E}\left[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right]}.$$

Proof. Fix Z and a design $(N, \mathcal{T}, \mathcal{P})$. By Proposition 1, under quadratic loss the optimal predictor satisfies

$$\hat{y}^*(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \mathbb{E}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})].$$

It therefore suffices to compute the posterior mean of $\boldsymbol{\beta}_\mathcal{T}$ given $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ in the Gaussian

linear model.
By Lemma 1, conditional on $\boldsymbol{X}_{N,\mathcal{T}}$ the training outcomes satisfy

$$\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}}\boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \mid \boldsymbol{X}_{N,\mathcal{T}} \sim \mathcal{N}\left(\boldsymbol{0}_N, \sigma_\varepsilon^2(t)\tau^2\, \boldsymbol{I}_N\right),$$

and the prior is $\boldsymbol{\beta}_{\mathcal{T}} \sim \mathcal{N}(\boldsymbol{0}_{|\mathcal{T}|}, \tau^2\boldsymbol{I}_{|\mathcal{T}|})$, independent of $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\varepsilon})$. This is the standard conjugate Gaussian regression model, so the posterior is Gaussian with precision matrix equal to the sum of prior and likelihood precisions:

$$\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})^{-1} = (\tau^2\boldsymbol{I}_{|\mathcal{T}|})^{-1} + (\sigma_\varepsilon^2(t)\tau^2)^{-1}\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}}.$$

Moreover, the posterior mean solves the normal equations for the posterior mode (which equals the mean in a Gaussian posterior):

$$\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)\mathbb{E}[\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}] = \boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{y}_N,$$

hence

$$\mathbb{E}[\boldsymbol{\beta}_{\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})] = \left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{y}_N.$$

Taking the subvector indexed by $\mathcal{P}\cap\mathcal{T}$ and plugging into the expression for $\hat{y}^*(\boldsymbol{D})$ yields the claimed formula. □

Proposition 4. *The value of a dataset $\boldsymbol{D}$ is*

$$v(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \underbrace{\left(\tau^2 \cdot \boldsymbol{X}_{N,\mathcal{T}}{}^\top\boldsymbol{X}_{N,\mathcal{T}}\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2 \cdot \boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}\right)_{\mathcal{P}\cap\mathcal{T}}}_{=\,\mathrm{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}] - \mathrm{Var}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}|(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})]}\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

Proof. By Proposition 4 (variance-reduction form), for any realized dataset $\boldsymbol{D}$,

$$v(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top\left(\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}) - \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right)\boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}.$$

Thus it suffices to compute the posterior covariance of $\boldsymbol{\beta}_{\mathcal{T}}$ in the Gaussian regression model of Lemma 1. With prior $\boldsymbol{\beta}_{\mathcal{T}} \sim \mathcal{N}(\boldsymbol{0}, \tau^2\boldsymbol{I}_{|\mathcal{T}|})$ and likelihood noise variance $\sigma_\varepsilon^2(t)\tau^2$, conjugacy yields

$$\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}) = \left((\tau^2\boldsymbol{I}_{|\mathcal{T}|})^{-1} + (\sigma_\varepsilon^2(t)\tau^2)^{-1}\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}}\right)^{-1} = \tau^2\sigma_\varepsilon^2(t)\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}.$$

Since $\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}}) = \tau^2\boldsymbol{I}_{|\mathcal{T}|}$, we obtain

$$\begin{aligned}
\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}}) - \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}) &= \tau^2\left[\boldsymbol{I}_{|\mathcal{T}|} - \sigma_\varepsilon^2(t)\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}\right] \\
&= \tau^2\left[\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right) - \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right]\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)^{-1} \\
&= \tau^2\,\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}}\left(\boldsymbol{X}_{N,\mathcal{T}}^\top\boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{|\mathcal{T}|}\right)^{-1}.
\end{aligned}$$

Taking the principal submatrix indexed by $\mathcal{P} \cap \mathcal{T}$ and plugging into the quadratic form above yields exactly the stated expression. $\square$

Proposition 5 (Closed Form Information)**.** *Under Assumption 1, the training information satisfies*

$$\mathcal{I}(n,t) = \frac{n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1}{2t}\left(1 - \sqrt{1 - \frac{4nt}{\left(n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1\right)^2}}\right),$$

Proof. Fix a sequence $\{\mathcal{D}(Z)\}$ satisfying Assumption 1, and write $T(Z) \equiv |\mathcal{T}(Z)|$. By definition and Gaussian conjugacy, the τ^2 at numerator and denominator cancel out and

$$\mathcal{I}(n,t) = 1 - \lim_{Z\to\infty} \frac{\sigma_\varepsilon^2(t)}{T(Z)} \mathbb{E}\left[\operatorname{tr}\left(\boldsymbol{X}_{N(Z),\mathcal{T}(Z)}^\top \boldsymbol{X}_{N(Z),\mathcal{T}(Z)} + \sigma_\varepsilon^2(t)\boldsymbol{I}_{T(Z)}\right)^{-1}\right].$$

Under the maintained Gaussian design, we can write $\boldsymbol{X}_{N(Z),\mathcal{T}(Z)} = \frac{\sigma_x}{\sqrt{Z}}\boldsymbol{G}_Z$ with $(\boldsymbol{G}_Z)_{ik} \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$, so that

$$\boldsymbol{X}^\top\boldsymbol{X} = \frac{\sigma_x^2}{Z}\boldsymbol{G}_Z^\top\boldsymbol{G}_Z = \frac{N(Z)}{Z}\Big(\frac{\sigma_x^2}{N(Z)}\boldsymbol{G}_Z^\top\boldsymbol{G}_Z\Big).$$

Let $S_Z \equiv \frac{1}{N(Z)}\boldsymbol{G}_Z^\top\boldsymbol{G}_Z$. Then $N(Z)/Z \to n$ and $T(Z)/N(Z) \to t/n$ by Assumption **??**, and

$$\frac{1}{T(Z)}\operatorname{tr}\left(\boldsymbol{X}^\top\boldsymbol{X} + \sigma_\varepsilon^2(t)\boldsymbol{I}\right)^{-1} = \frac{1}{n\sigma_x^2}\cdot\frac{1}{T(Z)}\operatorname{tr}\left(S_Z + a\boldsymbol{I}\right)^{-1} + o(1), \qquad a \equiv \frac{\sigma_\varepsilon^2(t)}{n\sigma_x^2} = \frac{1 - t + \frac{\sigma_\eta^2}{\sigma_x^2}}{n}.$$

By the Marchenko–Pastur theorem in the form of Bai–Silverstein's resolvent limit (see Marčenko and Pastur (1967) and Bai and Silverstein (2009)), the normalized resolvent trace $\frac{1}{T(Z)}\operatorname{tr}(S_Z + a\boldsymbol{I})^{-1}$ converges to the Stieltjes transform $m(-a)$ of the MP law with aspect ratio $\gamma \equiv t/n$. In particular, $m(-a)$ is the unique positive solution to

$$a\gamma\, m^2 + (a + 1 - \gamma)\, m - 1 = 0,$$

hence

$$m(-a) = \frac{-(a+1-\gamma) + \sqrt{(a+1-\gamma)^2 + 4a\gamma}}{2a\gamma}.$$

Combining the previous displays gives

$$\mathcal{I}(n,t) = 1 - a\, m(-a).$$

A short simplification yields

$$\mathcal{I}(n,t) = \frac{a+1+\gamma}{2\gamma}\left(1 - \sqrt{1 - \frac{1}{\gamma\left(\frac{a+1+\gamma}{2\gamma}\right)^2}}\right),$$

and substituting back $\gamma = \dfrac{t}{n}$ and $a = \dfrac{\sigma_\varepsilon^2(t)}{n\sigma_x^2}$ gives

$$\mathcal{I}(n,t) = \frac{n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1}{2t}\left(1 - \sqrt{1 - \frac{4nt}{\left(n + \frac{\sigma_\eta^2}{\sigma_x^2} + 1\right)^2}}\right),$$

as claimed. □

Theorem 1 (Asymptotic design value)**.** *Let $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ be a sequence of designs satisfying Assumption 1 with asymptotic design dimensions (n,t,p), with $n \geq 0$ and $1 \geq t \geq p \geq 0$. Then the asymptotic value exists and satisfies*

$$\mathcal{V}(n,t,p) = \underbrace{\sigma_x^2\, p}_{\text{usable target signal}} \times \underbrace{\tau^2\, \mathcal{I}(n,t)}_{\text{learning per trained coefficient}}.$$

where $\mathcal{I}(n,t)$ is the relative training information defined in Definition 3.

Proof of Theorem 1. Fix a sequence of designs $\{\mathcal{D}(Z)\}_{Z\in\mathbb{N}_+}$ satisfying Assumption 1 with

$$\frac{N(Z)}{Z} \to n, \qquad \frac{|\mathcal{T}(Z)|}{Z} \to t, \qquad \frac{|\mathcal{P}(Z)|}{Z} \to p, \qquad 1 \geq t \geq p \geq 0.$$

By Corollary 1 (no useless prediction covariates), wlog $\mathcal{P}(Z) \subseteq \mathcal{T}(Z)$, so $\mathcal{P}(Z) \cap \mathcal{T}(Z) = \mathcal{P}(Z)$.

Step 1 (expected value conditional on training data). In the high-dimensional Gaussian regression approximation, Proposition 4 gives that the value of a realized dataset $D = ((y_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}})$ can be written as

$$v(\boldsymbol{D}) = \boldsymbol{x}_{0,\mathcal{P}}^\top \left(\tau^2\, \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} (\boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\, I_{|\mathcal{T}|})^{-1}\right)_{\mathcal{P}} \boldsymbol{x}_{0,\mathcal{P}},$$

where $\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2$ and $(\cdot)_{\mathcal{P}}$ denotes the principal submatrix restricted to indices in $\mathcal{P}$.

Condition on the training data $(y_N, \boldsymbol{X}_{N,\mathcal{T}})$, hence on the matrix inside the quadratic form. Since $\boldsymbol{x}_{0,k} \sim \mathcal{N}(0, \sigma_x^2/Z)$ i.i.d. across k and independent of the training sample,

$$\mathbb{E}[v(\boldsymbol{D}) \mid y_N, \boldsymbol{X}_{N,\mathcal{T}}] = \operatorname{tr}\left(\left(\tau^2 \cdot \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} (\boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t) \cdot \boldsymbol{I})^{-1}\right)_{\mathcal{P}} \mathbb{E}[\boldsymbol{x}_{0,\mathcal{P}} \boldsymbol{x}_{0,\mathcal{P}}^\top]\right).$$

But $\mathbb{E}[\boldsymbol{x}_{0,\mathcal{P}} \boldsymbol{x}_{0,\mathcal{P}}^\top] = (\sigma_x^2/Z)\, I_{|\mathcal{P}|}$, hence

$$\mathbb{E}[v(\boldsymbol{D}) \mid y_N, \boldsymbol{X}_{N,\mathcal{T}}] = \frac{\sigma_x^2}{Z} \operatorname{tr}\left(\left(\tau^2 \cdot \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} (\boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t) \cdot \boldsymbol{I})^{-1}\right)_{\mathcal{P}}\right).$$

Step 2 (reduce the trace on $\mathcal{P}$ to a per-coordinate average). By index-wise symmetry (covariates i.i.d. across k) and $\mathcal{P}(Z) \subseteq \mathcal{T}(Z)$ with $|\mathcal{P}(Z)|/|\mathcal{T}(Z)| \to p/t$ when $t > 0$, the

average diagonal element of the matrix

$$\boldsymbol{M}_Z \;\equiv\; \tau^2 \cdot \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} (\boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t) \cdot \boldsymbol{I}_{|\mathcal{T}|})^{-1}$$

is the same across coordinates in $\mathcal{T}(Z)$ in expectation. Therefore,

$$\mathbb{E}[\mathrm{tr}((\boldsymbol{M}_Z)_{\mathcal{P}})] = \frac{|\mathcal{P}(Z)|}{|\mathcal{T}(Z)|} \mathbb{E}[\mathrm{tr}(\boldsymbol{M}_Z)] \,.$$

Taking expectations of Step 1 and using the preceding identity yields

$$\mathbb{E}[v(\boldsymbol{D})] = \frac{\sigma_x^2}{Z} \frac{|\mathcal{P}(Z)|}{|\mathcal{T}(Z)|} \mathbb{E}[\mathrm{tr}(\boldsymbol{M}_Z)] \,.$$

Step 3 (express $\mathbb{E}[\mathrm{tr}(\boldsymbol{M}_Z)]$ via training information). For ridge/Bayesian regression, the posterior covariance of $\beta_{\mathcal{T}}$ is

$$\mathrm{Var}(\beta_{\mathcal{T}} \mid y_N, \boldsymbol{X}_{N,\mathcal{T}}) = \tau^2 \cdot \left(\boldsymbol{I}_{|\mathcal{T}|} + \sigma_\varepsilon^{-2}(t) \cdot \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}}\right)^{-1}.$$

Using the identity $\boldsymbol{A}(\boldsymbol{A} + c \cdot \boldsymbol{I})^{-1} = \boldsymbol{I} - c \cdot (\boldsymbol{A} + c \cdot \boldsymbol{I})^{-1}$ with $\boldsymbol{A} = \boldsymbol{X}^\top \boldsymbol{X}$ and $c = \sigma_\varepsilon^2(t)$, one gets

$$\frac{1}{|\mathcal{T}|} \mathrm{tr}(\boldsymbol{M}_Z) = 1 - \frac{1}{\tau^2 |\mathcal{T}|} \mathrm{tr}(\mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}} \mid y_N, \boldsymbol{X}_{N,\mathcal{T}})) \,.$$

By Definition 3, along the proportional-growth sequence the limit of the right-hand side exists and equals $\mathcal{I}(n, t)$. Hence,

$$\lim_{Z \to \infty} \frac{1}{|\mathcal{T}(Z)|} \mathbb{E}[\mathrm{tr}(\boldsymbol{M}_Z)] = \mathcal{I}(n, t).$$

Step 4 (take limits). Combine Steps 2–3:

$$\mathbb{E}[v(\boldsymbol{D})] = \frac{\sigma_x^2}{Z} |\mathcal{P}(Z)| \left(\frac{1}{|\mathcal{T}(Z)|} \mathbb{E}[\mathrm{tr}(\boldsymbol{M}_Z)]\right).$$

As $Z \to \infty$, $|\mathcal{P}(Z)|/Z \to p$ and the parenthesis converges to $\mathcal{I}(n, t)$, so

$$\lim_{Z \to \infty} \mathbb{E}[v(\boldsymbol{D})] = \sigma_x^2 \, p \, \tau^2 \, \mathcal{I}(n, t).$$

By Definition 4, $\mathcal{V}(n, t, p) = \lim_{Z \to \infty} V(\mathcal{D}(Z)) = \lim_{Z \to \infty} \mathbb{E}[v(\boldsymbol{D})]$, which proves

$$\mathcal{V}(n, t, p) = \sigma_x^2 \, p \, \tau^2 \, \mathcal{I}(n, t),$$

as claimed. □

Proposition 6 (Characterization of the optimal design)**.** *Let (n^*, t^*, p^*) be a local optimum of*

$\Pi(n, t, p)$.

1. *If $n^* > 0$ and $0 < p^* < t^* < 1$, then*

$$C_N'(n^*) = \tau^2\sigma_x^2\, p^*\, \mathcal{I}_n(n^*, t^*), \tag{1}$$
$$C_{\mathcal{P}}'(p^*) = \tau^2\sigma_x^2\, \mathcal{I}(n^*, t^*), \tag{2}$$
$$C_{\mathcal{T}}'(t^*) = \tau^2\sigma_x^2\, p^*\, \mathcal{I}_t(n^*, t^*). \tag{3}$$

2. *If $p^* = t^* \in (0, 1)$ and $n^* > 0$, then*

$$C_N'(n^*) = \tau^2\sigma_x^2\, t^*\, \mathcal{I}_n(n^*, t^*), \tag{4}$$
$$C_{\mathcal{T}}'(t^*) + C_{\mathcal{P}}'(t^*) = \tau^2\sigma_x^2(\mathcal{I}(n^*, t^*) + t^*\, \mathcal{I}_t(n^*, t^*)). \tag{5}$$

Second-order conditions for local optimality are verified in the proof.

Proof. The Lagrangian of the design problem is

$$\mathcal{L} = \tau^2\sigma_x^2\, p\, \mathcal{I}(n, t) - C_N(n) - C_{\mathcal{T}}(t) - C_{\mathcal{P}}(p) + \lambda_n n + \lambda_p p + \lambda_{tp}(t - p) + \lambda_t(1 - t),$$

with multipliers $\lambda_n, \lambda_p, \lambda_{tp}, \lambda_t \geq 0$.

KKT conditions. Stationarity requires

$$\tau^2\sigma_x^2\, p\, \partial_n\mathcal{I}(n, t) - C_N'(n) + \lambda_n = 0, \tag{10}$$
$$\tau^2\sigma_x^2\, \mathcal{I}(n, t) - C_{\mathcal{P}}'(p) + \lambda_p - \lambda_{tp} = 0, \tag{11}$$
$$\tau^2\sigma_x^2\, p\, \partial_t\mathcal{I}(n, t) - C_{\mathcal{T}}'(t) + \lambda_{tp} - \lambda_t = 0. \tag{12}$$

Complementary slackness requires

$$\lambda_n n = 0, \quad \lambda_p p = 0, \quad \lambda_{tp}(t - p) = 0, \quad \lambda_t(1 - t) = 0,$$

together with feasibility $n \geq 0$ and $0 \leq p \leq t \leq 1$.

Interior case. If $n > 0$ and $0 < p < t < 1$, then $\lambda_n = \lambda_p = \lambda_{tp} = \lambda_t = 0$, so (10)–(12) reduce to the interior conditions in Proposition 6. Moreover, since $C_{\mathcal{T}}'(t) > 0$, interior optimality in t requires $\partial_t\mathcal{I}(n, t) \geq 0$.

Second-order conditions (interior). Assume $C_N, C_{\mathcal{T}}, C_{\mathcal{P}}$ are twice continuously differentiable on the relevant domains, and let (n, t, p) denote an interior candidate satisfying the interior KKT conditions. Let $\Pi(n, t, p) \equiv \tau^2\sigma_x^2\, p\, \mathcal{I}(n, t) - C_N(n) - C_{\mathcal{T}}(t) - C_{\mathcal{P}}(p)$, and write Π_{ab} for second

partial derivatives. The Hessian of Π at (n, t, p) is

$$H \equiv \begin{pmatrix} \Pi_{nn} & \Pi_{nt} & \Pi_{np} \\ \Pi_{nt} & \Pi_{tt} & \Pi_{tp} \\ \Pi_{np} & \Pi_{tp} & \Pi_{pp} \end{pmatrix} = \begin{pmatrix} \tau^2\sigma_x^2\, p\, \mathcal{I}_{nn} - C_N'' & \tau^2\sigma_x^2\, p\, \mathcal{I}_{nt} & \tau^2\sigma_x^2\, \mathcal{I}_n \\ \tau^2\sigma_x^2\, p\, \mathcal{I}_{nt} & \tau^2\sigma_x^2\, p\, \mathcal{I}_{tt} - C_{\mathcal{T}}'' & \tau^2\sigma_x^2\, \mathcal{I}_t \\ \tau^2\sigma_x^2\, \mathcal{I}_n & \tau^2\sigma_x^2\, \mathcal{I}_t & -C_{\mathcal{P}}'' \end{pmatrix},$$

where all derivatives of $\mathcal{I}$ and $C_{\cdot}$ are evaluated at (n, t, p). Since $C_{\mathcal{P}}''(p) > 0$ implies $\Pi_{pp} < 0$, a sufficient condition for (n, t, p) to be a *strict* local maximum is that the Schur complement of the (p, p) block is negative definite:

$$S \equiv \begin{pmatrix} \Pi_{nn} & \Pi_{nt} \\ \Pi_{nt} & \Pi_{tt} \end{pmatrix} - \frac{1}{\Pi_{pp}} \begin{pmatrix} \Pi_{np} \\ \Pi_{tp} \end{pmatrix} \begin{pmatrix} \Pi_{np} & \Pi_{tp} \end{pmatrix} \prec 0.$$

Equivalently, $S_{nn} < 0$, $S_{tt} < 0$, and $\det(S) > 0$.

Boundary case $p = t$. If $n > 0$ and $p = t \in (0, 1)$, then $\lambda_p = \lambda_t = 0$ while $\lambda_{tp} \geq 0$. Eliminating λ_{tp} from (11)–(12) yields

$$C_{\mathcal{T}}'(t) + C_{\mathcal{P}}'(t) = \tau^2\sigma_x^2\,(\mathcal{I}(n, t) + t\,\partial_t\mathcal{I}(n, t)),$$

which characterizes the optimal boundary scope, while (10) gives $C_N'(n) = \tau^2\sigma_x^2\, t\,\partial_n\mathcal{I}(n, t)$.

Second-order conditions (boundary $p = t$). Consider the restricted objective

$$\tilde{\Pi}(n, t) \equiv \Pi(n, t, t) = \tau^2\sigma_x^2\, t\,\mathcal{I}(n, t) - C_N(n) - C_{\mathcal{T}}(t) - C_{\mathcal{P}}(t).$$

A sufficient condition for (n, t) to be a *strict* local maximum of the restricted problem is that the Hessian of $\tilde{\Pi}$ be negative definite at (n, t), i.e.,

$$\tilde{H} \equiv \begin{pmatrix} \tilde{\Pi}_{nn} & \tilde{\Pi}_{nt} \\ \tilde{\Pi}_{nt} & \tilde{\Pi}_{tt} \end{pmatrix} \prec 0,$$

where

$$\tilde{\Pi}_{nn} = \tau^2\sigma_x^2\, t\,\mathcal{I}_{nn} - C_N'', \qquad \tilde{\Pi}_{nt} = \tau^2\sigma_x^2(\mathcal{I}_n + t\,\mathcal{I}_{nt}), \qquad \tilde{\Pi}_{tt} = \tau^2\sigma_x^2(2\mathcal{I}_t + t\,\mathcal{I}_{tt}) - C_{\mathcal{T}}'' - C_{\mathcal{P}}''.$$

Equivalently, $\tilde{\Pi}_{nn} < 0$, $\tilde{\Pi}_{tt} < 0$, and $\tilde{\Pi}_{nn}\tilde{\Pi}_{tt} - \tilde{\Pi}_{nt}^2 > 0$.

□